Markovian Anticipated BSDEs: Well-posedness, Comparison, Feynman-Kac Representation, and Approximation

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Abstract

In this work, we introduce and study a class of anticipated backward stochastic differential equations (ABSDEs) whose driver depends on the conditional distribution of a functional of the solution in the future. This framework unifies and extends previous works on ABSDEs, including those where the dependence occurs through conditional expectations or conditional risk measures. Under Lipschitz and square integrability assumptions in a Brownian filtration, we establish a priori estimates, existence and uniqueness of square integrable solutions, and a comparison principle under an additional assumption on the anticipated term. In the Markovian case, we prove that the anticipated term inherits the Markov structure and that the solution admits a Feynman-Kac representation in terms of a deterministic function of the underlying factor process. This value function solves a PDE nonlocal in both time and space, meant in the viscosity sense. Last, we show that an implicit ABSDE time-discretization scheme achieves the classical square-root convergence rate.

Keywords: Anticipated BSDEs, Conditional distribution, Feynman-Kac formula, Nonlocal PDEs, Derivative pricing, Risk Management.

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1 Introduction

Classical BSDEs

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s)ds - \int_t^T Z_s dW_s, \quad 0 \le t \le T, \tag{1}$$

where ξ is the terminal condition, f is the driver, and (Y, Z) the solution, are widely studied since the pioneer work of Pardoux and Peng (1990), see also El Karoui et al. (1997). Here, the driver only depends on the values of the solution at current time, making the equation "local in time". A key result is the Feynman-Kac representation: when the data ξ and f are Markovian, i.e. functions of an underlying Markov process X, the solution can be represented as $Y_t = u(t, X_t)$ where the value function u solves a semilinear parabolic partial differential equation.

With anticipated BSDEs (ABSDEs) such as introduced by Peng and Yang (2009), the driver at time t depends not only on the current value (Y_t, Z_t) of the solution, but also on the *conditional expectation* of the solution at future times. This creates a structure "nonlocal in time" that complicates the mathematical analysis. Due to the nonlocal nature of their driver, ABSDEs present several mathematical challenges compared to classical BSDEs, and their study has been pursued in various ways. Yang and Elliott (2013) consider an anticipated dependence in the sense of functional dependence with respect to the processes Y and Z in the future. Cheridito and Nam (2017) use abstract fixed-point arguments for studying a class of backward stochastic equations (BSEs) that extend classical BSDEs and include many examples of generalized BSDEs as well as semimartingale backward equation. Klimsiak and Rzymowski (2023) study BSDEs on a general filtration with drivers depending on the martingale part of the solution. Hu et al. (2026) consider multi-dimensional anticipated backward stochastic differential equations with quadratic growth.

ABSDEs arise in several applications. Peng and Yang (2009) initially introduced them for dealing with stochastic optimal control problems with delay. Further papers in this direction include Chen and Wu (2010) and Meng et al. (2025). On the financial side, regulated financial institutions must hold collateral and capital against their potential future losses. Basel Committee on Banking Supervision (2013) requires banks to compute their collateral and capital based on value-at-risk or expected shortfall of these future losses. On top of the traditional credit valuation adjustment (CVA) for counterparty default risk, the pricing of derivative instruments incorporates various valuation adjustments (XVAs) including margin valuation adjustment (MVA) for the cost of having to raise segregated collateral in the form of initial margin, funding valuation adjustment (FVA) for the cost of having to raise rehypothecable collateral in the form of variation margin, and capital valuation adjustment (KVA) for the cost of having to remunerate shareholders for their capital at risk. When accounted for in a dynamic XVA setup, collateral and capital loop lead to XVA ABSDEs. The MVA equation of Bourgey et al. (2020) is thus already an ABSDE. As reviewed in Crépey (2026, Chapter VII) building on Crépey et al. (2020), the fact that the KVA reserve is eligible to capital at risk, itself usable as variation margin, leads to a system of ABSDEs for the FVA and the KVA.

With these motivations in mind, we consider in this paper a class of ABSDEs, encompassing both Peng and Yang (2009) and Crépey et al. (2020), whose driver depends on the conditional distribution of a functional of the solution in the future. Relative to the above-mentioned works, whereas Yang and Elliott (2013) anticipate through a functional dependence on the future trajectory of (Y, Z), Cheridito and Nam (2017) and Klimsiak and Rzymowski (2023) through abstract fixed-point or martingale-part formulations, and Hu et al. (2026) through conditional expectations in a multi-dimensional quadratic setting, our driver depends on the future of the solution through its conditional distribution, which is precisely the structure required by the risk-measure applications described above.

Under Lipschitz conditions, in a semimartingale setup driven by Brownian motions, a careful analysis of the conditional distributions and fixed-point arguments yield a priori estimates, existence and uniqueness of the solution. We also provide explicit conditions granting comparison. In the Markovian case, we establish a Feynman-Kac representation. When comparison holds, the corresponding value function u is a viscosity solution to a partial differential equation, which is nonlocal, not only in space (as for classical BSDEs in jump-diffusion models or with McKean-Vlasov BSDEs) but also in time: the equation at point (t, x) involves the solution on the entire time-space domain beyond time t. We then prove the convergence, at a square-root rate of an implicit time-discretization ABSDE scheme. In our numerical experiments, space is handled by neural-network regressions in the spirit of Barrera et al. (2026), in line with the scheme of Abbas-Turki et al. (2025) (which was restricted to a dependence of f on the conditional law of the solution via conditional expected shortfalls, and deprived of mathematical convergence analysis); the convergence analysis of the ensuing space-discretization is left for future research.

The rest of the paper is organized as follows. Section 2 develops our fundamental ABSDE results including Proposition 2.4 for a priori estimates, Theorem 2.7 on existence and uniqueness of solutions, and the comparison principle of Theorem 2.8. Section 3 specializes these results to the Markovian case, Theorem 3.6 showing that $Y_t = u(t, X_t)$ then holds for some Hölder-regular value function u. We then establish in Section 4 the connection to the nonlocal partial differential equation (PDE) (27): Lemma 4.1 verifies that a smooth solution (if any) to (27) coincides with the value function u; conversely, Theorem 4.2 shows that, under conditions ensuring comparison for Markovian ABSDEs, the value function u is a viscosity solution to (27). Section 5 proves that the implicit time-discretization scheme for classical BSDEs extends to the anticipated framework, without any loss in convergence rate. This is illustrated numerically in Section **??**. Section 7 concludes the paper with a discussion of open questions and future research directions.

The main contributions with respect to previous literature are thus: on the mathematical side, a Feynman-Kac representation for Markovian ABSDEs, in terms of a value function u defined on the state-space of the factor process itself (no space-lifting required despite of the dependence of the coefficient on the conditional distribution of the solution in the future), but solving a PDE nonlocal in time and space; on the numerical side, a proof of convergence for the implicit Markovian ABSDEs time-discretization scheme of Abbas-Turki et al. (2025), at the square-root rate of classical BSDEs.

1.1 Notations

We denote by $|\cdot|$ the Euclidean norm of a vector (or matrix) in the dimension(s) of its argument. Given nonnegative constants T (a maturity) and θ (a terminal delay) and a filtered probability space $(\Omega, \mathcal{A}, \mathfrak{F} = (\mathfrak{F}_t)_{0\le t\le T+\theta}, \mathbb{P})$, where the filtration $\mathfrak{F}$ satisfies the usual conditions, we denote by $Prog$ the progressive sigma algebra on $\Omega \times [0,T]$ and we consider the following spaces of m-dimensional processes U, given positive integers $m, p \ge 1$:

$$\mathcal{H}^p(\mathfrak{F}, \mathbb{R}^m) = \left\{ \mathfrak{F} \text{ progressive processes } U\,;\, \|U\|^p_{\mathcal{H}^p} = \mathbb{E}\left[\int_0^{T+\theta} |U_t|^p dt\right] < \infty \right\},$$
$$\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m) = \left\{ \mathfrak{F} \text{ adapted continuous processes } U\,;\, \|U\|^2_{\mathcal{S}^2} = \mathbb{E}\left[\sup_{0\le t\le T+\theta} |U_t|^2\right] < \infty \right\}.$$

Note that $\|\cdot\|_{\mathcal{H}^2}$ and $\|\cdot\|_{\mathcal{S}^2}$ define respective norms on $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$ and $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$ and that, for any $\beta \ge 0$,

$$\|U\|_{\mathcal{H}^2_\beta} = \mathbb{E}\left[\int_0^{T+\theta} |U_t|^2 e^{\beta t} dt\right]^{\frac{1}{2}} \quad \text{and } \|U\|_{\mathcal{S}^2_\beta} = \mathbb{E}\left[\sup_{0\le t\le T+\theta} |U_t|^2 e^{\beta t}\right]^{\frac{1}{2}}$$

define norms respectively equivalent to $\|\cdot\|_{\mathcal{H}^2} \equiv \|\cdot\|_{\mathcal{H}^2_0}$ on $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$ and to $\|\cdot\|_{\mathcal{S}^2} \equiv \|\cdot\|_{\mathcal{S}^2_0}$ on $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$. For $\beta \ge 0$, $0 \le s \le t \le T+\theta$ and for $U \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^m)$, resp. $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^m)$, let also

$$\|U\|_{\mathcal{H}^2_\beta(s,t)} = \mathbb{E}\left[\int_s^t |U_r|^2 e^{\beta r} dr\right]^{\frac{1}{2}} \text{ and } \|U\|_{\mathcal{H}^2_\beta(t)} = \|U\|_{\mathcal{H}^2_\beta(0,t)}\,, \text{ resp.}$$
$$\|U\|_{\mathcal{S}^2_\beta(s,t)} = \mathbb{E}\left[\sup_{s\le r\le t} |U_r|^2 e^{\beta r}\right]^{\frac{1}{2}} \text{ and } \|U\|_{\mathcal{S}^2_\beta(t)} = \|U\|_{\mathcal{S}^2_\beta(0,t)}.$$

Given a Polish space (E, d_E), we denote by $\mathcal{B}(E)$ the Borel sigma-algebra on E and by $\mathcal{P}(E)$ the set of probability measures on $(E, \mathcal{B}(E))$. For $p \ge 1$, we denote by $\mathcal{P}_p(E)$ the subset of probability measures $\mu \in \mathcal{P}(E)$ with finite pth moment, i.e. such that $\int_E d_E(e, e_0)^p \mu(de) < \infty$ holds for any fixed $e_0 \in E$. We endow $\mathcal{P}_p(E)$ with the p-Wasserstein distance defined by

$$W_p(\mu, \mu') = \left(\inf_{\pi \in \Pi(\mu,\mu')} \int_{E\times E} d_E(e, e')^p \pi(de, de')\right)^{\frac{1}{p}},\ \mu, \mu' \in \mathcal{P}_p(E),$$

where $\Pi(\mu, \mu')$ is the set of probability measures on $E \times E$ with marginals μ and μ'. By Villani (2009, Theorem 6.18), $(\mathcal{P}_p(E), W_p)$ is a Polish space. We denote by $T_\sharp \mu$ the pushforward (aka image) measure $\mu \circ (T^{-1})$ of a measure μ by a map T.

We write a.s. and a.e. for $d\mathbb{P}$ almost surely and $dt \otimes d\mathbb{P}$ almost everywhere (on $[0, T+\theta] \times \Omega$, unless otherwise stated). For a random variable ε defined on $(\Omega, \mathcal{A}, \mathbb{P})$ and taking values in a Polish space E, we denote by $\mathbb{L}(\varepsilon)$ its law, i.e. the probability measure on $(E, \mathcal{B}(E))$ defined by $\mathbb{L}(\varepsilon)(B) = \mathbb{P}(\varepsilon \in B)$, $B \in \mathcal{B}(E)$. For a sub-sigma-algebra $\mathcal{F}$

of $\mathcal{A}$, we denote by $\mathbb{L}(\varepsilon|\mathcal{F}) = \mathbb{P}(\varepsilon \in de|\mathcal{F})$ the conditional law of ε given $\mathcal{F}$, i.e. the probability measure on $\mathcal{B}(E)$ defined by $\mathbb{L}(\varepsilon|\mathcal{F})(B) = \mathbb{P}(\varepsilon \in B|\mathcal{F})$, $B \in \mathcal{B}(E)$. For any $t \leq T+\theta$, we denote by $\mathbb{P}_t(A)$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional probability of an event $A \in \mathcal{A}$, by $\mathbb{E}_t[\xi]$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional expectation of an integrable real random variable ξ, and by $\mathbb{L}_t(\xi)$ the $(\mathfrak{F}_t, \mathbb{P})$ conditional law of an E valued $\mathcal{A}$ measurable random variable ξ, where E is Polish.

We denote by C (or $C(\rho)$) a generic positive constant (modulo perhaps dependence on parameters ρ but only on ρ) that may change from line to line.

2 BSDEs Anticipated in Conditional Distribution

After defining the anticipated BSDEs (ABSDEs) of interest in this work, we establish foundational results: a priori estimates, existence, uniqueness, and comparison of solutions.

Hereafter $\mathfrak{F}$ is assumed to be the natural augmented filtration of a q variate Brownian motion $W = (W_t)_{0\leq t\leq T+\theta}$, for some positive integer q. We also denote by $\mathfrak{F}^T = (\mathfrak{F}_t)_{t\in[0,T]}$ the restriction of $\mathfrak{F}$ to $[0,T]$, for some maturity $T > 0$ and terminal delay $\theta \geq 0$.

Given a positive integer n , we consider the n variate ABSDE

$$\begin{aligned} &Z_t = \Psi_t,\ t \in (T, T+\theta] \text{ and, for } t \in [0, T+\theta], \\ &Y_t = \Phi_{t\vee T} + \int_{t\wedge T}^{T} f(s, Y_s, Z_s, \mu_s^{Y,Z})ds - \int_{t\wedge T}^{T} Z_s dW_s, \end{aligned} \tag{2}$$

where

- (Φ, Ψ) is an $\mathbb{R}^n \times \mathbb{R}^{n\times q}$ valued $\mathfrak{F}$ adapted process,
- $f : \Omega\times[0,T]\times\mathbb{R}^n\times\mathbb{R}^{n\times q}\times\mathcal{P}_2(E) \to \mathbb{R}^n$ is a $\mathcal{P}\otimes\mathcal{B}^n\otimes\mathcal{B}^{n\times q}\otimes\mathcal{B}(\mathcal{P}_2(E))$ measurable map, where E is a Polish space,
- $\mu : \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q}) \to \mathcal{H}^0(\mathfrak{F}, \mathcal{P}_2(E)), (t, \omega, Y, Z) \mapsto \mu_t^{Y,Z}(\omega)$ with E a fixed Polish space and $\mathcal{H}^0(\mathfrak{F}, \mathcal{P}_2(E))$ the set of $\mathcal{P}_2(E)$ valued progressive processes, represents the anticipated term in the sense that, given $(Y, Z) \in \mathcal{H}^2(\mathfrak{F}; \mathbb{R}^n \times \mathbb{R}^{n\times q})$ and $t \in [0, T]$, the $\mathcal{P}_2(E)$ valued variable $\mu_t^{Y,Z}$ depends on the process at future times $s \in [t, T+\theta]$ but only via its $\mathfrak{F}_t$ conditional distribution (see Assumption 2.1(iii) and Remark 2.3 below).

Remark 2.1. Even if the terminal delay θ is equal to 0, at any time $t < T$, the anticipated term $\mu_t^{Y,Z}$ still depends on the future (from t to T) of the solution, so the equation is still an anticipated BSDE. However, as t approaches T, the influence of the future path diminishes as the time-to-maturity $T - t$ decreases to 0. If $\theta > 0$, the influence of the future path remains significant up to the maturity.

We will say that (f, Φ, Ψ, μ) are *ABSDE standard parameters* if they satisfy the following:

Assumption 2.1. **(i)** $(\Phi_t)_{T\le t\le T+\theta}$ is continuous, $(\Psi_t)_{T\le t\le T+\theta}$ is progressive, and

$$\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi_s|^2+\int_T^{T+\theta}|\Psi_s|^2ds\right]<\infty.$$

(ii) There exists a real constant Λ such that for any $(t,y,y',z,z',\mu,\mu')\in[0,T]\times(\mathbb{R}^n)^2\times(\mathbb{R}^{n\times q})^2\times\mathcal{P}_2(E)^2$,

$$\big|f(t,y,z,\mu)-f(t,y',z',\mu')\big|\le\Lambda\big(|y-y'|+|z-z'|+W_2(\mu,\mu')\big),\ \mathbb{P}\text{-a.s.}.$$

In addition, $f(\cdot,0,0,\mu_\cdot^{0,0})\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)$.
(iii) For any $(Y,Z),(Y',Z')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$ and any $0\le t\le T$, we have

$$W_2(\mu_t^{Y,Z},\mu_t^{Y',Z'})^2\le\mathbb{E}_t\left[\int_t^{T+\theta}\big(|Y_s-Y'_s|^2+|Z_s-Z'_s|^2\big)\rho_t(ds)\right],\ \text{a.s.},\tag{3}$$

for some family of nonnegative Borel measures $\rho_t, t\in[0,T]$, respectively supported on $[t,T+\theta]$, with bounded mass $\sup_{t\in[0,T]}\rho_t([t,T+\theta])<\infty$.

In the rest of the paper, to simplify the exposition, we assume that $\rho_t(ds)=ds$ in (3).

Remark 2.2. The results of the paper extend to the case of general measures ρ_t in (3), the constants in the estimates then depending on $\bar R=\sup_{t\in[0,T]}\rho_t([t,T+\theta])$ and on a bound C_ν on the Lebesgue density of the superposition measure $\nu(ds)=\int_0^T\rho_t(ds)\,dt$ on $[0,T+\theta]$ (assumed to exist). Indeed, Assumption 2.1(iii) is only ever used after integration against $e^{\beta t}dt$ on $[0,T]$, and, as $e^{\beta t}\le e^{\beta s}$ on the support of ρ_t, the corresponding terms are bounded via the Fubini theorem by integrals against $e^{\beta s}\nu(ds)\le C_\nu e^{\beta s}ds$. In particular, the seminal setting of Example 2.4(i), for which $\rho_t=\delta_{t+\theta}$, hence $\bar R=1$ and $C_\nu\le1$, is covered.

The assumptions on Φ,Ψ and f are quite standard in BSDE setups.

Remark 2.3. If (Y,Z) and (Y',Z') are such that $(Y,Z)=(Y',Z')$ a.e. on $[t,T+\theta]$, then under Assumption 2.1(iii), $\mu_t^{Y,Z}=\mu_t^{Y',Z'}$ a.s.. The values of (Y,Z) before t are then irrelevant to compute $\mu_t^{Y,Z}$, motivating the anticipated terminology.

Assumption 2.1(iii) is satisfied for a wide class of anticipated terms, as the following lemma (see also the examples below) shows.

Lemma 2.1. *Assume that* $\widetilde\mu_t^{Y,Z}(\omega)=\mathbb{L}(A(t,Y,Z)|\mathfrak{F}_t)(\omega)$ *for some map* $A:[0,T]\times\mathcal{H}^2(\mathfrak{F};\mathbb{R}^n\times\mathbb{R}^{n\times q})\to L^2(\mathfrak{F}_{T+\theta};E)$ *such that for each* $t\in[0,T]$ *and each* $(Y,Z),(Y',Z')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})^2$, $(t,\omega)\mapsto A(t,Y,Z)(\omega)$ *is jointly measurable, and*

$$\mathbb{E}_t\left[d_E(A(t,Y,Z),A(t,Y',Z'))^2\right]\le\mathbb{E}_t\left[\int_t^{T+\theta}\big(|Y_s-Y'_s|^2+|Z_s-Z'_s|^2\big)ds\right],\ a.s..\tag{4}$$

Then there exists $\mu:\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})\to\mathcal{H}^0(\mathfrak{F},\mathcal{P}_2(E))$ *satisfying Assumption 2.1(iii) and such that, for each* $(Y,Z)\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, $\mu^{Y,Z}$ *is a (progressive) version of* $\widetilde\mu^{Y,Z}$.

Proof. We first fix $(Y,Z) \in \mathcal{H}^2(\mathfrak{F};\mathbb{R}^n \times \mathbb{R}^{n\times q})$ and show that $\widetilde{\mu}^{Y,Z}$ admits a progressive version.

As E is a Polish space, the space $\mathcal{P}_2(E)$ equipped with the topology of the Wasserstein distance W_2 is also a Polish space. Hence, there exists a countable family $\{f_n\}_{n\geq 1} \subset C_b(E)$ that separates measures in $\mathcal{P}_2(E)$, that is, $\nu = \nu'$ if and only if $\int_E f_n \, d\nu = \int_E f_n \, d\nu'$ for all $n \geq 1$.

Consider the injective map $\mathcal{I} : \mathcal{P}_2(E) \to \mathbb{R}^{\mathbb{N}}$ defined by $\mathcal{I}(\nu) = \left(\int_E f_n \, d\nu\right)_{n\geq 1}$. As W_2-convergence implies weak convergence and each f_n is bounded and continuous, $\mathcal{I}$ is continuous and thus Borel measurable. By the Lusin-Souslin Theorem on injective Borel maps (see (Kechris, 2012, Theorem 15.1)), the image $\mathcal{P}' := \mathcal{I}(\mathcal{P}_2(E))$ is a Borel subset of $\mathbb{R}^{\mathbb{N}}$, and the inverse map $\mathcal{I}^{-1} : \mathcal{P}' \to \mathcal{P}_2(E)$ is Borel measurable.

For each $n \geq 1$, the real valued process $(f_n(A(t,Y,Z)))_{t\geq 0}$ is bounded and jointly measurable. By the Optional Projection Theorem, there exists an optional (hence progressive) real-valued process $\nu^n = (\nu^n_t)_{t\geq 0}$ such that for each stopping time $\tau \leq T$,

$$\nu^n_\tau = \mathbb{E}[f_n(A(\tau,Y,Z)) \mid \mathcal{F}_\tau] \quad \text{a.s. on } \{\tau \leq T\}.$$

We then define the $\mathbb{R}^{\mathbb{N}}$ valued progressively measurable process $\nu_t = (\nu^n_t)_{n\geq 1}$, and observe that, for each stopping time $\tau \leq T$, $\nu_\tau = \mathcal{I}(\mathbb{L}(A(\tau,Y,Z)|\mathfrak{F}_\tau)) \in \mathcal{P}'$ almost surely.

We now show that $N = \{(t,\omega) \in [0,T] \times \Omega : \nu_t(\omega) \notin \mathcal{P}'\}$ is evanescent, i.e., its projection $\pi_\Omega(N) := \{\omega \in \Omega : \exists t \leq T \text{ s.t. } \nu_t(\omega) \notin \mathcal{P}'\}$ has zero probability. Since ν is an optional process and $\mathcal{P}'$ is a Borel set, N is an optional subset of $[0,T]\times\Omega$. Assuming by contradiction that $\mathbb{P}(\pi_\Omega(N)) > 0$, the Optional Section Theorem guarantees the existence of a stopping time τ such that $\{(\tau(\omega),\omega) : \tau(\omega) \leq T\} \subset N$ and $\mathbb{P}(\tau \leq T) \geq \frac{1}{2}\mathbb{P}(\pi_\Omega(N)) > 0$. This implies that $\nu_\tau \notin \mathcal{P}'$ on $\{\tau \leq T\}$, which is a contradiction since $\nu_\tau = \mathcal{I}(\mathbb{L}(A(\tau,Y,Z)|\mathfrak{F}_\tau)) \in \mathcal{P}'$ almost surely. Therefore, N is evanescent.

We now define $\widehat{\nu}_t = \nu_t \mathbf{1}_{N^c} + \mathcal{I}(\delta_{x_0})\mathbf{1}_N$, where $x_0 \in E$ is an arbitrary fixed point. Then $\widehat{\nu}$ is an optional (hence progressive) process taking values exclusively in $\mathcal{P}'$, and for each time $t \leq T$, $\widehat{\nu}_t = \nu_t = \mathcal{I}(\mu^{Y,Z}_t)$ almost surely. Therefore, $\widehat{\nu}$ is a progressive modification of $\mathcal{I}(\mu^{Y,Z})$.

To conclude, we define the $\mathcal{P}_2(E)$ valued process $\mu^{Y,Z}_t = \mathcal{I}^{-1}(\widehat{\nu}_t)$. Since $\mathcal{I}^{-1}$ is Borel measurable and $\widehat{\nu}$ is progressive, $\mu^{Y,Z}$ is progressive. Furthermore, for each $t \leq T$, $\mathcal{I}(\mu^{Y,Z}_t) = \widehat{\nu}_t = \nu_t = \mathcal{I}(\widetilde{\mu}^{Y,Z}_t)$ almost surely. By the injectivity of $\mathcal{I}$, $\widetilde{\mu}^{Y,Z}_t = \mu^{Y,Z}_t$ almost surely, demonstrating that $\mu^{Y,Z}$ is indeed a progressive modification of $\widetilde{\mu}^{Y,Z}$.

Last, (3) is an immediate consequence of (4), as for each $t \in [0,T]$,

$$W_2(\mu^{Y,Z}_t, \mu^{Y',Z'}_t)^2 = W_2(\widetilde{\mu}^{Y,Z}_t, \widetilde{\mu}^{Y',Z'}_t)^2 \leq \mathbb{E}_t\left[d_E(A(t,Y,Z), A(t,Y',Z'))^2\right] \quad \text{a.s.},$$

which proves the lemma. □

Example 2.4. *(i) Classical anticipated BSDEs (Peng and Yang (2009)):* $E = \mathbb{R}^n \times \mathbb{R}^{n\times q}$, $\mu^{Y,Z}_t := \mathbb{L}_t(Y_{t+\theta}, Z_{t+\theta})$ for some $\delta \geq 0$, and $f(t,y,z,\mu) = \widetilde{f}(t,y,z,\int y\mu(dy,dz), \int z\mu(dy,dz))$. In this case, the anticipated term depends on the conditional distribution of the solution at a single future time $t+\theta$. This is the original ABSDE framework of Peng and Yang (2009), where the driver depends on $\mathbb{E}_t[Y_{t+\theta}]$ and $\mathbb{E}_t[Z_{t+\theta}]$.

Regarding Assumption 2.1(iii), Lemma 2.1 provides a progressive modification of $\mu^{Y,Z}$, and for any $(Y,Z),(Y',Z') \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n \times \mathbb{R}^{n\times q})$ and any $0 \le t \le T$, we almost surely have

$$W_2(\mu_t^{Y,Z},\mu_t^{Y',Z'})^2 \le \mathbb{E}_t\left[\left(\left|Y_{t+\theta}-Y'_{t+\theta}\right|^2+\left|Z_{t+\theta}-Z'_{t+\theta}\right|^2\right)\right]$$
$$= \mathbb{E}_t\left[\int_t^{T+\theta}\left(\left|Y_s-Y'_s\right|^2+\left|Z_s-Z'_s\right|^2\right)\rho_t(ds)\right],$$

where $\rho_t(ds)=\delta_{t+\theta}(ds)$ is the Dirac measure at $t+\theta$, with mass 1.
(ii) Expected shortfall and XVA applications: $n=2$, $\mu_t^{Y,Z} := \mathbb{L}_t(\int_t^{t+\theta} Z_s dW_s)$, and $f(t,y,z,\mu)=\widetilde{f}(t,y,z,\mathbb{ES}_\alpha(\mu))$ for some $\alpha \in (0,1)$, where $\mathbb{ES}_\alpha(\mu)=\frac{\int_{\mathbb{Va}\mathbb{R}_\alpha(\mu)}^{\infty} x\mu(dx)}{\int_{\mathbb{Va}\mathbb{R}_\alpha(\mu)}^{\infty}\mu(dx)}$ is the Expected Shortfall of μ at level α, and $\mathbb{Va}\mathbb{R}_\alpha(\mu)=\inf\{x\in\mathbb{R}:\mu((-\infty,x])\ge\alpha\}$ is the quantile of μ at level α. The capital at risk of an investment bank can be modeled as an expected shortfall of its future trading gains/losses, with law μ. The variation margin is a tracker of the mark-to-market of a bank's derivative portfolio, needing to be posted by the bank as collateral to its clients. As explained in Crépey (2026, Chapter VII), the fungibility between variation margin and capital at risk leads to an ABSDE of the above form (extended to a jump-diffusion setup encompassing the defaults of the clients of the bank) for the bank's FVA and KVA, i.e. the costs for the bank of funding its variation margin and of remunerating its shareholders for their capital at risk.

Regarding Assumption 2.1(iii), Lemma 2.1 provides a progressive modification of $\mu^{Y,Z}$, and note that for any $(Y,Z),(Y',Z') \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n \times \mathbb{R}^{n\times q})$ and any $0 \le t \le T$, we almost surely have

$$W_2(\mu_t^{Y,Z},\mu_t^{Y',Z'})^2 \le \mathbb{E}_t\left[\left|\int_t^{t+\theta} Z_s dW_s-\int_t^{t+\theta} Z'_s dW_s\right|^2\right]$$
$$= \mathbb{E}_t\left[\int_t^{t+\theta}\left|Z_s-Z'_s\right|^2 ds\right] = \mathbb{E}_t\left[\int_t^{T+\theta}\left|Z_s-Z'_s\right|^2\rho_t(ds)\right],$$

where $\rho_t(ds)$ is the Lebesgue measure on $[t,t+\theta]$, with mass θ.

2.1 A Priori Estimates

Lemma 2.2. *Let f and μ satisfy Assumptions 2.1(ii)-(iii). Then, for any $\beta \ge 0$ and any $(Y,Z),(Y',Z'),(\widetilde{Y},\widetilde{Z}),(\widetilde{Y}',\widetilde{Z}') \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, we have*

$$\|f(\cdot,Y_\cdot,Z_\cdot,\mu_\cdot^{\widetilde{Y},\widetilde{Z}})-f(\cdot,Y'_\cdot,Z'_\cdot,\mu_\cdot^{\widetilde{Y}',\widetilde{Z}'})\|^2_{\mathcal{H}^2_\beta(T)}$$
$$\le 3\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta(T)}+\|Z-Z'\|^2_{\mathcal{H}^2_\beta(T)}+T\|\widetilde{Y}-\widetilde{Y}'\|^2_{\mathcal{H}^2_\beta}+T\|\widetilde{Z}-\widetilde{Z}'\|^2_{\mathcal{H}^2_\beta}\right).$$

In particular,

$$\|f(\cdot,Y_\cdot,Z_\cdot,\mu_\cdot^{Y,Z})-f(\cdot,Y'_\cdot,Z'_\cdot,\mu_\cdot^{Y',Z'})\|^2_{\mathcal{H}^2_\beta(T)} \le$$
$$3(T+1)\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta}+\|Z-Z'\|^2_{\mathcal{H}^2_\beta}\right).$$

Proof. Using Assumption 2.1(ii)-(iii), we compute

$$\begin{aligned}
&\mathbb{E}\left[\int_0^T e^{\beta t}\left|f(t,Y_t,Z_t,\mu_t^{\widetilde{Y},\widetilde{Z}})-f(t,Y'_t,Z'_t,\mu_t^{\widetilde{Y}',\widetilde{Z}'})\right|^2 dt\right]\\
&\qquad\le 3\Lambda^2\mathbb{E}\left[\int_0^T e^{\beta t}\left(|Y_t-Y'_t|^2+|Z_t-Z'_t|^2+W_2(\mu_t^{\widetilde{Y},\widetilde{Z}},\mu_t^{\widetilde{Y}',\widetilde{Z}'})^2\right)dt\right]\\
&=3\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta(T)}+\|Z-Z'\|^2_{\mathcal{H}^2_\beta(T)}+\mathbb{E}\left[\int_0^T e^{\beta t}W_2(\mu_t^{\widetilde{Y},\widetilde{Z}},\mu_t^{\widetilde{Y}',\widetilde{Z}'})^2dt\right]\right) \qquad (5)\\
&\le 3\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta(T)}+\|Z-Z'\|^2_{\mathcal{H}^2_\beta(T)}+\right.\\
&\quad\left.\mathbb{E}\left[\int_0^T e^{\beta t}\mathbb{E}_t\left[\int_t^{T+\theta}\left|\widetilde{Y}_s-\widetilde{Y}'_s\right|^2+\left|\widetilde{Z}_s-\widetilde{Z}'_s\right|^2 ds\right]dt\right]\right)\\
&\le 3\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta(T)}+\|Z-Z'\|^2_{\mathcal{H}^2_\beta(T)}+T\|\widetilde{Y}-\widetilde{Y}'\|^2_{\mathcal{H}^2_\beta}+T\|\widetilde{Z}-\widetilde{Z}'\|^2_{\mathcal{H}^2_\beta}\right).
\end{aligned}$$

□

Proposition 2.3. *For (f,Φ,Ψ,μ) satisfying Assumption 2.1, let (Y,Z) be an $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ solution to the ABSDE (2). Then $Y\in\mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)$.*

Proof. By (2) and the triangular inequality, we have

$$\sup_{0\le t\le T}|Y_t|^2\le 3|\Phi_T|^2+3T\int_0^T\left|f\left(t,Y_t,Z_t,\mu_t^{Y,Z}\right)\right|^2dt+3\sup_{0\le t\le T}\left|\int_t^T Z_s dW_s\right|^2.$$

First, by Assumption 2.1(i), one has $\mathbb{E}\left[|\Phi_T|^2\right]\le\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi_s|^2\right]<\infty$.
Second, using Assumption 2.1(ii) and Lemma 2.2 with $\beta=0$, we compute

$$\begin{aligned}
&\mathbb{E}\left[\int_0^T\left|f(t,Y_t,Z_t,\mu_t^{Y,Z})\right|^2dt\right]\\
&\le 2\mathbb{E}\left[\int_0^T\left|f(t,Y_t,Z_t,\mu_t^{Y,Z})-f(t,0,0,\mu_t^{0,0})\right|^2dt\right]+2\mathbb{E}\left[\int_0^T\left|f(t,0,0,\mu_t^{0,0})\right|^2dt\right] \qquad (6)\\
&\le 6(T+1)\Lambda^2\|(Y,Z)\|^2_{\mathcal{H}^2}+2\|f(\cdot,0,0,\mu^{0,0})\|^2_{\mathcal{H}^2(T)}.
\end{aligned}$$

Third, the Burkholder-Davis-Gundy inequality of Karatzas and Shreve (1991, Theorem 3.28) yields

$$\begin{aligned}
\mathbb{E}\left[\sup_{0\le t\le T}\left|\int_t^T Z_s dW_s\right|^2\right]&\le 2\mathbb{E}\left[\left|\int_0^T Z_s dW_s\right|^2\right]+2\mathbb{E}\left[\sup_{0\le s\le T}\left|\int_0^s Z_r dW_r\right|^2\right]\\
&\le 10\mathbb{E}\left[\int_0^T|Z_s|^2ds\right]<\infty.
\end{aligned}$$

All in all, this proves that $\|Y\|^2_{\mathcal{S}^2(T)}<\infty$.
Last, $\|Y\|^2_{\mathcal{S}^2}\le\|Y\|^2_{\mathcal{S}^2(T)}+\|\Phi\|^2_{\mathcal{S}^2(T,T+\theta)}<\infty$, by Assumption 2.1(i), which proves that $Y\in\mathcal{S}^2(\mathfrak{F},\mathbb{R}^n)$, as Y is continuous by (2). □

Proposition 2.4. *For $i = 1, 2$, let (Y^i, Z^i) be an $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ solution to the ABSDE (2) with parameters $(f^i, \Phi^i, \Psi^i, \mu^i)$ satisfying Assumption 2.1 for some Lipschitz constant Λ^i of f^i. We set $\mu_t^i = \mu_t^{i,Y^i,Z^i}$, $i = 1, 2$, and $\mu_t^{1,2} = \mu_t^{1,Y^2,Z^2}$, $t \in [0, T]$ and we define, for any $0 \le t \le T + \theta$,*

$$\delta Y_t = Y_t^1 - Y_t^2, \delta Z_t = Z_t^1 - Z_t^2, \delta\mu_t = W_2(\mu_t^1, \mu_t^{1,2}), \delta\Phi_t = \Phi_t^1 - \Phi_t^2, \delta\Psi_t = \Psi_t^1 - \Psi_t^2,$$
$$\delta_2 f_t = f^1(t, Y_t^2, Z_t^2, \mu_t^{1,2}) - f^2(t, Y_t^2, Z_t^2, \mu_t^2).$$

Then, for any $\nu, \lambda, \eta, \beta > 0$ such that

$$\Lambda^1(2 + \lambda^2 + \nu^2 + \frac{T}{\nu^2}) + \eta^2 \le \beta \text{ and } \Lambda^1(\frac{1}{\lambda^2} + \frac{T}{\nu^2}) < 1, \tag{7}$$

we have

$$\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} \le T(1+\theta)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{T}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + T\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} \tag{8}$$

$$\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} \le \tag{9}$$
$$\frac{\nu^2\lambda^2}{\nu^2\lambda^2 - \Lambda^1(\nu^2 + T\lambda^2)}\left((1+\theta)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}\right).$$

In addition, there exists a real constant K only depending on (T, θ, Λ^1) such that

$$\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} \le C\left(\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}\right). \tag{10}$$

Proof. For $t \in [0, T]$, applying the Itô formula to $|\delta Y_t|^2$ and the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned}
&e^{\beta t}|\delta Y_t|^2 + \beta\int_t^T e^{\beta s}|\delta Y_s|^2 ds + \int_t^T e^{\beta s}|\delta Z_s|^2 ds \\
=&e^{\beta T}|\delta Y_T|^2 + 2\int_t^T e^{\beta s}\langle \delta Y_s, f^1(s, Y_s^1, Z_s^1, \mu_s^1) - f^2(s, Y_s^2, Z_s^2, \mu_s^2)\rangle ds \\
&- 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
\le&e^{\beta T}|\delta\Phi_T|^2 + 2\int_t^T e^{\beta s}|\delta Y_s||f^1(s, Y_s^1, Z_s^1, \mu_s^1) - f^2(s, Y_s^2, Z_s^2, \mu_s^2)| ds \\
&- 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
\le&e^{\beta T}|\delta\Phi_T|^2 + 2\int_t^T e^{\beta s}|\delta Y_s||f^1(s, Y_s^1, Z_s^1, \mu_s^1) - f^1(s, Y_s^2, Z_s^2, \mu_s^{1,2})| ds \\
&+ 2\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s| ds - 2\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s.
\end{aligned} \tag{11}$$

We first prove (8)-(9). By Proposition 2.3, $\delta Y \in \mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n)$; by assumption, $\delta Z \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$; hence $(e^{\beta s}\delta Y_s^\top \delta Z_s)_{s\in[0,T]} \in \mathcal{H}^1(\mathfrak{F}, \mathbb{R}^q)$. In particular, for any $t \le T$,

$\int_t^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s$ is integrable with zero expectation. Hence

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \beta\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + \|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \leq \\
&\qquad \mathbb{E}\left[e^{\beta T}|\delta\Phi_T|^2\right] + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||f^1(s,Y_s^1,Z_s^1,\mu_s^1) - f^1(s,Y_s^2,Z_s^2,\mu_s^{1,2})|ds\right] \\
&\qquad\qquad + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s|ds\right].
\end{aligned}$$

Using Assumption 2.1(ii) on f^1 and the Young inequality, we obtain

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \beta\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + \|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \\
\leq&\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 2\Lambda^1\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s|\left(|\delta Y_s| + |\delta Z_s| + \delta\mu_s\right)ds\right] + 2\mathbb{E}\left[\int_t^T e^{\beta s}|\delta Y_s||\delta_2 f_s|ds\right] \\
\leq&\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \left(\Lambda^1\left(2+\lambda^2+\nu^2\right)+\eta^2\right)\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} \qquad (12)\\
&+ \frac{\Lambda^1}{\lambda^2}\|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} + \frac{\Lambda^1}{\nu^2}\|\delta\mu\|^2_{\mathcal{H}^2_\beta(t,T)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(t,T)}.
\end{aligned}$$

From Assumption 2.1(iii), using $e^{\beta s} \leq e^{\beta r}$ for $s \leq r$, we also have

$$\begin{aligned}
&\|\delta\mu\|^2_{\mathcal{H}^2_\beta(t,T)} \\
\leq&\mathbb{E}\left[\int_t^T e^{\beta s}\left(\int_T^{T+\theta}\left(|\delta\Phi_r|^2 + |\delta\Psi_r|^2\right)dr + \int_s^T\left(|\delta Y_r|^2 + |\delta Z_r|^2\right)dr\right)ds\right] \\
\leq&T\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} + T\|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} + T\theta\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + T\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}. \qquad (13)
\end{aligned}$$

Combining (12) and (13), we obtain

$$\begin{aligned}
&\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] + \left(\beta - \Lambda^1\left(2+\lambda^2+\nu^2+\frac{T}{\nu^2}\right) - \eta^2\right)\|\delta Y\|^2_{\mathcal{H}^2_\beta(t,T)} \\
&\qquad + \left(1 - \Lambda^1\left(\frac{1}{\lambda^2} + \frac{T}{\nu^2}\right)\right)\|\delta Z\|^2_{\mathcal{H}^2_\beta(t,T)} \\
&\leq \left(1 + \frac{\Lambda^1 T\theta}{\nu^2}\right)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{\Lambda^1 T}{\nu^2}\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)}. \qquad (14)
\end{aligned}$$

This implies, for (λ,η,ν,β) satisfying (7),

$$\mathbb{E}\left[e^{\beta t}|\delta Y_t|^2\right] \leq (1+\theta)\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + \frac{1}{\eta^2}\|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)},$$

as in particular $\Lambda^1\frac{T}{\nu^2} < 1$. Integrating over time, we obtain (8). Using (7) again, (14) (at $t=0$) implies (9).

To prove (10), starting from (11), we obtain

$$\begin{aligned}
e^{\beta t}|\delta Y_t|^2 \leq& e^{\beta T}|\delta\Phi_T|^2 + 2\int_0^T e^{\beta s}|\delta Y_s|^2 ds + \int_0^T e^{\beta s}|f^1(s,Y_s^1,Z_s^1,\mu_s^1) - f^1(s,Y_s^2,Z_s^2,\mu_s^{1,2})|^2 ds \\
&+ \int_0^T e^{\beta s}|\delta_2 f_s|^2 ds + 2\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|.
\end{aligned}$$

As the right hand side does not depend on t, taking the supremum over t and expectations yields

$$\begin{aligned}\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} &\leq \|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 2\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} \\ &\quad + \mathbb{E}\left[\int_0^T e^{\beta s}|f^1(s,Y^1_s,Z^1_s,\mu^1_s) - f^1(s,Y^2_s,Z^2_s,\mu^{1,2}_s)|^2 ds\right] \\ &\quad + \|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right].\end{aligned}$$

From Lemma 2.2, as $\|\delta Y\|^2_{\mathcal{S}^2_\beta} \leq \|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} + \|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)}$ and $\|\delta Z\|^2_{\mathcal{H}^2_\beta} = \|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} + \|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)}$, we obtain

$$\begin{aligned}\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} &\leq (1+3(T+1))\|\delta\Phi\|^2_{\mathcal{S}^2_\beta(T,T+\theta)} + 3(T+1)\|\delta\Psi\|^2_{\mathcal{H}^2_\beta(T,T+\theta)} \\ &\quad + (2+3(T+1)\Lambda^2)\|\delta Y\|^2_{\mathcal{H}^2_\beta(T)} + 3(T+1)\Lambda^2\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)} \\ &\quad + \|\delta_2 f\|^2_{\mathcal{H}^2_\beta(T)} + 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right].\end{aligned} \tag{15}$$

Moreover, the Burkholder-Davis-Gundy and Young inequalities imply the existence of some $C>0$ such that

$$\begin{aligned}&\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_s^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \leq \mathbb{E}\left[\left|\int_0^T e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \\ &\qquad + \mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_0^s e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \\ &\leq 2\mathbb{E}\left[\sup_{0\leq s\leq T}\left|\int_0^s e^{\beta r}\delta Y_r^\top \delta Z_r dW_r\right|\right] \leq 2C\mathbb{E}\left[\left(\int_0^T e^{2\beta r}|\delta Y_r|^2|\delta Z_r|^2 dr\right)^{\frac{1}{2}}\right] \\ &\leq 2C\mathbb{E}\left[\left(\sup_{0\leq r\leq T} e^{\beta r}|\delta Y_r|^2\right)^{\frac{1}{2}}\left(\int_0^T e^{\beta r}|\delta Z_r|^2 dr\right)^{\frac{1}{2}}\right] \leq \frac{1}{4}\|\delta Y\|^2_{\mathcal{S}^2_\beta(T)} + 4C^2\|\delta Z\|^2_{\mathcal{H}^2_\beta(T)}.\end{aligned} \tag{16}$$

All in all, plugging (16) into (15) and using (8)-(9), we obtain (10). □

From these a priori estimates, we immediately deduce some regularity properties.

Corollary 2.5. *Let $(f^\alpha,\Phi^\alpha,\Psi^\alpha,\mu^\alpha,\alpha\in\mathbb{R})$ be a family of parameters satisfying Assumption 2.1 with Lipschitz constant Λ independent of α. Assume that, for each $\alpha\in\mathbb{R}$, there exists an $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ solution (Y^α,Z^α) to the ABSDE (2) with parameters $(f^\alpha,\Phi^\alpha,\Psi^\alpha,\mu^\alpha)$. Assume in addition that, as $\alpha\to\alpha_0\in\mathbb{R}$, we have*

$$\begin{aligned}&\left\|f^\alpha\left(\cdot,Y^{\alpha_0},Z^{\alpha_0},\mu^{\alpha,Y^{\alpha_0},Z^{\alpha_0}}\right) - f^{\alpha_0}\left(\cdot,Y^{\alpha_0},Z^{\alpha_0},\mu^{\alpha_0,Y^{\alpha_0},Z^{\alpha_0}}\right)\right\|_{\mathcal{H}^2(T)}\to 0, \\ &\|\Phi^\alpha-\Phi^{\alpha_0}\|_{\mathcal{S}^2(T,T+\theta)}\to 0, \text{ and } \|\Psi^\alpha-\Psi^{\alpha_0}\|_{\mathcal{H}^2(T,T+\theta)}\to 0.\end{aligned}$$

Then the function $\alpha\to(Y^\alpha,Z^\alpha)$ from $\mathbb{R}$ to $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$ is continuous.

Proof. This is an immediate consequence of the a priori estimates (8)–(9) applied with $(f^1, \Phi^1, \Psi^1, \mu^1) = (f^\alpha, \Phi^\alpha, \Psi^\alpha, \mu^\alpha)$ and $(f^2, \Phi^2, \Psi^2, \mu^2) = (f^{\alpha_0}, \Phi^{\alpha_0}, \Psi^{\alpha_0}, \mu^{\alpha_0})$, noting that $\delta_2 f$, $\delta\Phi$ and $\delta\Psi$ all converge to 0 as $\alpha \to \alpha_0$ by assumption. □
Note that the function from $\mathbb{R}$ to $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ is continuous too, as we have proved that $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ solutions actually belong to $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$, and (9)-(10) controls their differences in $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$.

2.2 Existence and Uniqueness

Lemma 2.6. *Let (f, Φ, Ψ, μ) satisfy Assumption 2.1. For any $(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we consider the classical BSDE with terminal condition Φ_T and driver $(\omega, t, y, z) \mapsto f(\omega, t, y, z, \mu_t^{\widetilde{Y},\widetilde{Z}}(\omega))$. This classical BSDE admits a unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution.*

By definition, the driver $(\omega, t, y, z) \mapsto f(\omega, t, y, z, \mu_t^{\widetilde{Y},\widetilde{Z}}(\omega))$ is $Prog \otimes \mathcal{B}(\mathbb{R}^n) \otimes \mathcal{B}(\mathbb{R}^{n\times q})$ measurable, where $Prog$ is the progressive sigma-algebra associated with $\mathfrak{F}$.
Under Assumption 2.1, we have $\mathbb{E}\left[\Phi_T^2\right] \le \|\Phi\|^2_{\mathcal{S}^2} < \infty$, for $(\omega, t, y, y', z, z') \in \Omega \times [0,T] \times (\mathbb{R}^n)^2 \times (\mathbb{R}^{n\times q})^2$, $|f(\omega, t, y, z, \mu_t^{\widetilde{Y},\widetilde{Z}}(\omega)) - f(\omega, t, y', z', \mu_t^{\widetilde{Y},\widetilde{Z}}(\omega))| \le \Lambda(|y - y'| + |z - z'|)$, and,

$$
\begin{aligned}
&\|f(t, 0, 0, \mu_t^{\widetilde{Y},\widetilde{Z}})\|^2_{\mathcal{H}^2(T)} \\
&\quad \le \|f(\cdot, 0, 0, \mu^{\widetilde{Y},\widetilde{Z}}) - f(\cdot, 0, 0, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} + \|f(\cdot, 0, 0, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} \\
&\quad \le \Lambda^2 \|W_2(\mu^{\widetilde{Y},\widetilde{Z}}, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} + \|f(\cdot, 0, 0, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} \\
&\quad \le \mathbb{E}\left[\int_0^T \mathbb{E}_t\left[\int_t^{T+\theta} (\|\widetilde{Y}_s\|^2 + \left|\widetilde{Z}_s\right|^2) ds\right] dt\right] + \|f(\cdot, 0, 0, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} \\
&\quad \le T\|(\widetilde{Y}, \widetilde{Z})\|^2_{\mathcal{H}^2} + \|f(\cdot, 0, 0, \mu^{0,0})\|^2_{\mathcal{H}^2(T)} < \infty.
\end{aligned}
$$

Hence the classical BSDE with terminal condition Φ_T and driver $f(\omega, t, y, z, \mu_t^{\widetilde{Y},\widetilde{Z}}(\omega))$ admits a unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution, by El Karoui et al. (1997, Theorem 2.1). □
For any $(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$, we define $\Gamma(\widetilde{Y}, \widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ as this unique $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ solution on $[0, T]$ and as (Φ, Ψ) on $[T, T + \theta]$.
Note that, since $Y_{|[0,T]} \in \mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$, Φ is continuous on $[T, T + \theta]$ and satisfies $\mathbb{E}\left[\sup_{T\le t\le T+\theta} |\Phi_t|^2\right] < \infty$, and $(Y_{|[0,T]})_T = \Phi_T$, the first component of $\Gamma(\widetilde{Y}, \widetilde{Z})$ is in $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n)$.

The main result of this section is:

Theorem 2.7. *Let (f, Φ, Ψ, μ) satisfy Assumption 2.1.*
(i) *The ABSDE (2) admits a unique $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^{n\times q})$ solution (Y, Z), which can be characterized as the unique fixed point of the contraction Γ of $\mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$.*
(ii) *The sequence of Picard iterates defined by $(Y^0, Z^0) = (0, 0)$ (or, more generally, by any $(Y^0, Z^0) \in \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^n \times \mathbb{R}^{n\times q})$) and, for $k \ge 0$, $(Y^{k+1}, Z^{k+1}) = \Gamma(Y^k, Z^k)$, converges to (Y, Z) in $\mathcal{H}^2(\mathfrak{F}^T, \mathbb{R}^n \times \mathbb{R}^{n\times q})$ and $d\mathbb{P} \otimes ds$ a.s.. The sequence $(Y^k)_k$ also converges to Y in $\mathcal{S}^2(\mathfrak{F}^T, \mathbb{R}^n)$.*

Proof. (i) We define the closed subset

$$\mathcal{H}^2_{\Phi,\Psi} = \{(\widetilde{Y},\widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n \times \mathbb{R}^{n\times q}); (\widetilde{Y},\widetilde{Z}) = (\Phi,\Psi) \text{ on } (T,T+\theta]\},$$

which is a complete metric space under $\|\cdot\|_\beta$. By construction, Γ maps $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$ into $\mathcal{H}^2_{\Phi,\Psi}$, and in particular $\mathcal{H}^2_{\Phi,\Psi}$ into itself. We now prove that Γ is a contraction of $(\mathcal{H}^2_{\Phi,\Psi}, \|\cdot\|_\beta)$, for β large enough, which will be enough to obtain the existence and uniqueness of an $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$ solution to (2). Then, by Proposition 2.3, we obtain that $(Y,Z) \in \mathcal{S}^2(\mathfrak{F},\mathbb{R}^n) \times \mathcal{H}^2(\mathfrak{F},\mathbb{R}^{n\times q})$, proving (i).

For $(\widetilde{Y},\widetilde{Z}),(\widetilde{Y}',\widetilde{Z}') \in \mathcal{H}^2_{\Phi,\Psi}$, we set $(Y,Z) = \Gamma(\widetilde{Y},\widetilde{Z})$, $(Y',Z') = \Gamma(\widetilde{Y}',\widetilde{Z}')$, $\delta Y = Y - Y'$, $\delta Z = Z - Z'$, $\delta\widetilde{Y} = \widetilde{Y} - \widetilde{Y}'$ and $\delta\widetilde{Z} = \widetilde{Z} - \widetilde{Z}'$. Note that $\delta\widetilde{Y} = \delta\widetilde{Z} = 0$ on $(T,T+\theta]$, so that in particular $\|(\delta\widetilde{Y},\delta\widetilde{Z})\|_{\mathcal{H}^2_\beta} = \|(\delta\widetilde{Y},\delta\widetilde{Z})\|_{\mathcal{H}^2_\beta(T)}$. By the Itô formula, the Cauchy-Schwarz inequality, and the Young inequality, we have, as also $Y_T = Y'_T = \Phi_T$ (cf. the beginning of the proof of Proposition 2.4),

$$\begin{aligned}
&|\delta Y_0|^2 + \int_0^T e^{\beta s}\left(\beta|\delta Y_s|^2 + |\delta Z_s|^2\right) ds \\
&= 2\int_0^T e^{\beta s}\delta Y_s^\top \big(f(s,Y_s,Z_s,\mu^{\widetilde{Y},\widetilde{Z}}) - f(s,Y'_s,Z'_s,\mu^{\widetilde{Y}',\widetilde{Z}'})\big)ds - 2\int_0^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s \\
&\le \int_0^T e^{\beta s}(\frac{\beta}{2}|\delta Y_s|^2 + \frac{2}{\beta}|f(s,Y_s,Z_s,\mu^{\widetilde{Y},\widetilde{Z}}) - f(s,Y'_s,Z'_s,\mu^{\widetilde{Y}',\widetilde{Z}'})|^2)ds \\
&\qquad - 2\int_0^T e^{\beta s}\delta Y_s^\top \delta Z_s dW_s.
\end{aligned}$$

Hence we obtain by Assumption 2.1, with Λ the Lipschitz constant of f,

$$\begin{aligned}
&\mathbb{E}\left[\int_0^T e^{\beta s}(\frac{\beta}{2}|\delta Y_s|^2 + |\delta Z_s|^2)ds\right] \\
&\le \frac{2}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}|f(s,Y_s,Z_s,\mu^{\widetilde{Y},\widetilde{Z}}) - f(s,Y'_s,Z'_s,\mu^{\widetilde{Y}',\widetilde{Z}'})|^2 ds\right] \\
&\le \frac{6\Lambda^2}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}(|\delta Y_s|^2 + |\delta Z_s|^2)ds\right] + \frac{6T\Lambda^2}{\beta}\mathbb{E}\left[\int_0^T e^{\beta s}(|\delta\widetilde{Y}_s|^2 + |\delta\widetilde{Z}_s|^2)ds\right].
\end{aligned}$$

For $\beta > 2 \vee \Lambda^2(12T+6)$, this implies

$$\|(\delta Y,\delta Z)\|^2_{\mathcal{H}^2_\beta(T)} \le \frac{6T\Lambda^2}{\beta - 6\Lambda^2}\|(\delta\widetilde{Y},\delta\widetilde{Z})\|^2_{\mathcal{H}^2_\beta(T)} \le \frac{1}{2}\|(\delta\widetilde{Y},\delta\widetilde{Z})\|^2_{\mathcal{H}^2_\beta(T)}, \tag{17}$$

which shows that Γ is a contraction of $\mathcal{H}^2_{\Phi,\Psi}$, hence (2) admits a unique $\mathcal{H}^2_{\Phi,\Psi}$ solution, and in particular a unique $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$ solution.
(ii) Since Γ maps $\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$ into $\mathcal{H}^2_{\Phi,\Psi}$, all Picard iterates (Y^k,Z^k) belong to $\mathcal{H}^2_{\Phi,\Psi}$ for $k \ge 1$, and the convergence in $\mathcal{H}^2(\mathfrak{F}^T,\mathbb{R}^n\times\mathbb{R}^{n\times q})$ is guaranteed by the contraction property. In particular, for every $\varepsilon > 0$, we obtain by the Markov inequality

and (17),

$$\begin{aligned}\sum_{k\geq 0}(\mathbb{P}\otimes ds)\left(e^{\beta s}\left(|Y_s^{k+1}-Y_s^k|^2+|Z_s^{k+1}-Z_s^k|^2\right)\geq\varepsilon\right)&\\ \leq\frac{1}{\varepsilon}\sum_{k\geq 0}\|(Y^{k+1}-Y^k,Z^{k+1}-Z^k)\|^2_{\mathcal{H}^2_\beta}&\\ \leq\frac{\|(Y^1-Y^0,Z^1-Z^0)\|^2_{\mathcal{H}^2_\beta}}{\varepsilon}\sum_{k\geq 0}\frac{1}{2^k}<\infty.&\end{aligned}$$

Applying the Borel-Cantelli lemma, we eventually get

$$\limsup_k |Y_s^{k+1}-Y_s^k|^2+|Z_s^{k+1}-Z_s^k|^2=0,\ d\mathbb{P}\otimes ds \text{ a.s..}$$

In particular, for $d\mathbb{P}\otimes ds$ almost every $(\omega,t)\in\Omega\times[0,T+\theta]$, $(Y_t^k(\omega))$ and $(Z_t^k(\omega))_{k\geq 0}$ are Cauchy sequences, hence they converge, respectively to $Y_t(\omega)$ and $Z_t(\omega)$ (as these are already the limits in some square integrable sense, hence a subsequence of (Y^k,Z^k) is already almost surely converging to (Y,Z)).

To prove the convergence of $(Y^k)_k$ in $\mathcal{S}^2(\mathfrak{F}^T,\mathbb{R}^n)$, we apply the estimate (10) to the two classical BSDEs defining $(Y^{k+1},Z^{k+1})=\Gamma(Y^k,Z^k)$ and $(Y^k,Z^k)=\Gamma(Y^{k-1},Z^{k-1})$. As the terminal conditions coincide, this yields, for some constant $C\geq 0$ independent of k,

$$\begin{aligned}\|Y^{k+1}-Y^k\|^2_{\mathcal{S}^2_\beta(T)}&\leq C\|f(\cdot,Y^k,Z^k,\mu^{Y^k,Z^k})-f(\cdot,Y^k,Z^k,\mu^{Y^{k-1},Z^{k-1}})\|^2_{\mathcal{H}^2_\beta(T)}\\ &\leq CT\Lambda^2\|(Y^k-Y^{k-1},Z^k-Z^{k-1})\|^2_{\mathcal{H}^2_\beta(T)}\\ &\leq CT\Lambda^2 2^{-k+1}\|(Y^1-Y^0,Z^1-Z^0)\|^2_{\mathcal{H}^2_\beta(T)},\end{aligned}$$

where we used Assumption 2.1 and (17). Hence $\sum_{k\geq 1}\|Y^{k+1}-Y^k\|_{\mathcal{S}^2_\beta(T)}<\infty$, so $(Y^k)_k$ is Cauchy in the Banach space $\mathcal{S}^2(\mathfrak{F}^T,\mathbb{R}^n)$. Since already $Y^k\to Y$ in $\mathcal{H}^2(\mathfrak{F}^T,\mathbb{R}^n)$, this gives $Y^k\to Y$ in $\mathcal{S}^2(\mathfrak{F}^T,\mathbb{R}^n)$. □

2.3 Comparison

In this section, we consider the case of unidimensional ABSDEs, i.e. $n=1$, and we prove a comparison theorem. Toward this end, we will make use of the following stronger assumption regarding the "anticipated" part of the BSDE, namely, that the BSDE does not anticipate on the Z process. In other words, the measure $\mu^{Y,Z}$ only depends on Y.

Assumption 2.2. $\mu^{Y,Z}$ does not depend on Z: For any $(t,Y,Z)\in[0,T]\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}\times\mathbb{R}^q)$, we have $\mu_t^{Y,Z}=\widetilde{\mu}_t^Y$ for some $\widetilde{\mu}:[0,T]\times\Omega\times\mathcal{H}^2(\mathfrak{F},\mathbb{R})\to\mathcal{P}_2(E)$.

Remark 2.5. The process Ψ becomes thus irrelevant. We then say that $(f,\Phi,\widetilde{\mu})$ are standard ABSDE parameters when they satisfy Assumption 2.1. All the previous

estimates remain valid, with bounds that do not anymore depend on the future of Z or on Ψ. In particular, for any $(Y,Z),(Y',Z') \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}\times\mathbb{R}^q)$ and $\widetilde{Y},\widetilde{Y}' \in \mathcal{H}^2(\mathfrak{F},\mathbb{R})$, Assumption 2.1(iii) and Lemma 2.2 yield

$$W_2(\widetilde{\mu}_t^Y,\widetilde{\mu}_t^{Y'})^2 \le \mathbb{E}_t\left[\int_t^{T+\theta}|Y_s-Y'_s|^2ds\right] \text{ a.s., and}$$
$$\|f(\cdot,Y,Z,\widetilde{\mu}^{\widetilde{Y}}) - f(\cdot,Y',Z',\widetilde{\mu}^{\widetilde{Y}'})\|^2_{\mathcal{H}^2_\beta(T)}$$
$$\le 3\Lambda^2\left(\|Y-Y'\|^2_{\mathcal{H}^2_\beta(T)} + \|Z-Z'\|^2_{\mathcal{H}^2_\beta(T)} + T\|\widetilde{Y}-\widetilde{Y}'\|^2_{\mathcal{H}^2_\beta}\right).$$

We will also make use of the following monotonicity assumption.

Assumption 2.3. For any $(t,y,z)\in[0,T]\times\mathbb{R}\times\mathbb{R}^q$ and all $(Y,Y')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R})^2$, if $Y\ge Y'$ holds $dt\otimes\mathbb{P}$ almost surely on $[0,T+\theta]$, then, $\mathbb{P}$ almost surely,

$$f(\omega,t,y,z,\widetilde{\mu}_t^Y(\omega)) \ge f(\omega,t,y,z,\widetilde{\mu}_t^{Y'}(\omega)).$$

Theorem 2.8. *Let (f^a,Φ^a,μ^a) and (f^b,Φ^b,μ^b) be two ABSDE standard parameters satisfying Assumptions 2.1 and 2.2, with respective $\mathcal{S}^2(\mathfrak{F},\mathbb{R})\times\mathcal{H}^2(\mathfrak{F},\mathbb{R}^q)$ solutions (Y^a,Z^a) and (Y^b,Z^b) to the corresponding ABSDEs (2). We suppose that (f^a,μ^a) satisfies Assumption 2.3. If, for any $(y,z)\in\mathbb{R}\times\mathbb{R}^q$ and $\widetilde{Y}\in\mathcal{H}^2(\mathfrak{F},\mathbb{R})$,*

$$\begin{aligned} &f^b(\cdot,y,z,\mu^{b,\widetilde{Y}}) \ge f^a(\cdot,y,z,\mu^{a,\widetilde{Y}}), \text{ a.e. on } [0,T], \text{ and} \\ &\Phi^b\ge\Phi^a, dt\otimes\mathbb{P} \text{ a.e. on } [T,T+\theta], \end{aligned} \tag{18}$$

then

$$Y_t^b \ge Y_t^a \text{ a.s., for all } t\in[0,T+\theta].$$

Proof. We denote by Γ the contraction of the Banach space $(\mathcal{H}^2(\mathfrak{F},\mathbb{R}\times\mathbb{R}^q),\|\cdot\|_{\mathcal{H}^2_\beta})$, for β large enough, defined in Theorem 2.7(i) and associated to parameters (f^a,Φ^a,μ^a), so (Y^a,Z^a) is its unique fixed point. Note that Γ in fact does not depend on z, as the anticipated part of the BSDE does not depend on Z, by Assumption 2.2.

We define $(Y^1,Z^1)=\Gamma(Y^b,Z^b)$ and, for $k\ge1$, $(Y^{k+1},Z^{k+1})=\Gamma(Y^k,Z^k)$, so that the sequence $(Y^k,Z^k)_{k\ge1}$ converges to (Y^a,Z^a) in $\mathcal{H}^2(\mathfrak{F}^T,\mathbb{R}\times\mathbb{R}^q)$.

Supposing (18), we now prove that $Y^b\ge Y^k\ge Y^{k+1}$ for any $k\ge1$, which will be enough to prove that $Y^b\ge Y^a$, as $Y^k\to_{k\to\infty}Y^a$ a.e., by Theorem 2.7(ii) (applied with starting point $(Y^0,Z^0)=(Y^b,Z^b)$).

First, (Y^1,Z^1) (resp. (Y^b,Z^b)) is, on $[0,T]$, the unique $\mathcal{H}^2(\mathfrak{F}^T,\mathbb{R}\times\mathbb{R}^q)$ solution to the classical BSDE with terminal condition Φ_T^a (resp. Φ_T^b) and driver $f^a(\cdot,\mu^{a,Y^b})$ (resp. $f^b(\cdot,\mu^{b,Y^b})$). By (18), we have $\Phi_T^a\le\Phi_T^b$ and $f^a(\cdot,\mu^{a,Y^b})\le f^b(\cdot,\mu^{b,Y^b})$, hence the classical comparison theorem for BSDEs (see e.g. El Karoui et al. (1997, Theorem 2.2)) gives $Y_t^b\ge Y_t^1$ almost surely for any $t\in[0,T]$. In addition, on $[T,T+\theta]$, $Y^b=\Phi^b\ge\Phi^a=Y^1$, hence $Y^b\ge Y^1$ on $[0,T+\theta]$.

For $k=1$, Y^1 and Y^2 have the same terminal condition Φ_T^a and the respective drivers $f^a(\cdot,\mu^{a,Y^b})$ and $f^a(\cdot,\mu^{a,Y^1})$. By Assumption 2.3 on (f^a,μ^a), as $Y^b\ge Y^1$, we have $f^a(\cdot,\mu^{a,Y^b})\ge f^a(\cdot,\mu^{a,Y^1})$ so the classical comparison theorem for BSDEs gives

$Y_t^1 \geq Y_t^2$ a.s., for any $t \in [0,T]$, and hence for $t \in [0,T+\theta]$, as $Y^1 = \Phi^a = Y^2$ on $[T,T+\theta]$. Now, let us prove by induction that $Y^k \geq Y^{k+1}$ for any $k \geq 1$. Note that Y^{k+1} and Y^{k+2} have both terminal condition Φ_T^a and respective drivers $f^a(\cdot,\mu_\cdot^{a,Y^k})$ and $f^a(\cdot,\mu_\cdot^{a,Y^{k+1}})$. If $Y^k \geq Y^{k+1}$ holds by induction hypothesis, then Assumption 2.3 implies that $f^a(\cdot,\mu_\cdot^{a,Y^k}) \geq f^a(\cdot,\mu_\cdot^{a,Y^{k+1}})$, hence $Y_t^{k+1} \geq Y_t^{k+2}$ a.s., for any $t \in [0,T]$ (hence all $t \in [0,T+\theta]$) by the classical BSDEs comparison theorem.

We conclude that $Y_t^b \geq Y_t^k$ a.s., for any $t \in [0,T+\theta]$ and $k \geq 1$, and the monotone sequence $(Y^k)_{k\geq 1}$ converges to Y a.e., giving $Y_t^b \geq Y_t^a$ a.s., for any $t \in [0,T+\theta]$. Since Y^b and Y^a belong to $\mathcal{S}^2(\mathfrak{F},\mathbb{R})$ by Proposition 2.3, their trajectories are continuous. Hence, $Y_t^b(\omega) \geq Y_t^a(\omega)$ for a.e. t and a.s. ω extends by continuity to all $t \in [0,T+\theta]$, $\mathbb{P}$ almost surely. $\square$

3 Markovian ABSDEs

The Markovian case, with all randomness in the data driven by an underlying Markovian factor process X, deserves special attention for several reasons. First, it allows representing the solution (Y,Z) of the BSDE as deterministic functions of the time and the underlying forward process X, which will be used for proving the convergence of the time-discretization scheme in Section 5. Second, this deterministic representation enables us to obtain a Feynman-Kac ABSDE representation $Y_t = u(t,X_t)$, where the value function u solves a nonlocal partial differential equation. Finally, the Markovian framework is the typical setting for applications, where the anticipated features only depend on the current state of the underlying system and not on the past history of randomness.

The key ABSDE technical challenge in the Markovian case is to establish that the anticipated term inherits the Markov property. This requires showing that conditional distributions of functionals of future paths can be expressed as deterministic (measure valued) functions of the current state, despite their apparent dependence on future randomness. Once this representation is established, the classical machinery of viscosity solutions can be adapted to handle the nonlocal nature of the resulting PDE.

We consider the following Markovian ABSDE setup: For $(t,x) \in [0,T+\theta] \times \mathbb{R}^d$,

$$\begin{aligned} X_s &= x + \int_t^{s\vee t} b(r,X_r)dr + \int_t^{s\vee t} \sigma(r,X_r)dW_r, \quad s \in [0,T+\theta], \\ Y_s &= \Phi(s\vee T, X_{s\vee T}) + \int_{s\wedge T}^T f(r,X_r,Y_r,Z_r,\mu_r^{Y,Z})dr - \int_{s\wedge T}^T Z_r dW_r, \quad s \in [0,T+\theta], \\ Z_s &= \Psi(s,X_{s\vee T}), \quad s \in (T,T+\theta], \end{aligned} \tag{19}$$

for measurable maps $(b,\sigma) : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^d \times \mathbb{R}^{d\times q}$, $(\Phi,\Psi) : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^n \times \mathbb{R}^{n\times q}$, $f : [0,T]\times\mathbb{R}^d\times\mathbb{R}^n\times\mathbb{R}^{n\times q}\times\mathcal{P}_2(E) \to \mathbb{R}^n$ and $\mu : \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q}) \to \mathcal{H}^0(\mathfrak{F},\mathcal{P}_2(E))$, where $d \geq 1$ is an integer.

We say that the coefficients are standard Markovian parameters if they satisfy the following:

Assumption 3.1. **(i)** The maps Φ and Ψ are continuous, such that

$$\mathbb{E}\left[\sup_{T\le t\le T+\theta}|\Phi(t,0)|^2+\int_T^{T+\theta}|\Psi(t,0)|^2dt\right]<\infty,$$

and there exists a real constant Λ such that, for any $(t,x,x')\in[T,T+\theta]\times\mathbb{R}^d\times\mathbb{R}^d$,

$$\big|\Phi(t,x)-\Phi(t,x')\big|+\big|\Psi(t,x)-\Psi(t,x')\big|\le\Lambda|x-x'|.$$

Last, there exists a measurable map $\widetilde{\Psi}:[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^{n\times d}$ such that $\Psi=\widetilde{\Psi}^\top\sigma$ holds on $[T,T+\theta]$.
(ii) For any $(t,x,x',y,y',z,z',\mu,\mu')\in[0,T]\times(\mathbb{R}^d)^2\times(\mathbb{R}^n)^2\times(\mathbb{R}^{n\times q})^2\times\mathcal{P}_2(E)^2$, we have

$$\big|f(t,x,y,z,\mu)-f(t,x',y',z',\mu')\big|\le\Lambda(|x-x'|+|y-y'|+|z-z'|+W_2(\mu,\mu')).$$

In addition, $f(\cdot,0,0,0,\mu^{0,0})\in\mathcal{H}^2(\mathfrak{F},\mathbb{R}^n)$.
(iii) Assumption 2.1(iii) is in force.
(iv) The maps b and σ are continuous on $[0,T+\theta]\times\mathbb{R}^d$ and, for any $(t,x,x')\in[0,T+\theta]\times(\mathbb{R}^d)^2$,

$$\begin{aligned}&\big|b(t,x)-b(t,x')\big|+\big|\sigma(t,x)-\sigma(t,x')\big|\le\Lambda|x-x'|,\\&|b(t,x)|+|\sigma(t,x)|\le\Lambda(1+|x|).\end{aligned}$$

We denote by $\mathfrak{F}^t=(\mathfrak{F}^t_s)_{0\le s\le T+\theta}$ the filtration defined by $\mathfrak{F}^t_s=\sigma(W_r-W_t,t\le r\le t\vee s)$, $0\le s\le T+\theta$. Under Assumption 3.1(iv), Touzi (2012, Theorems 2.2 and 2.4) show that $X^{t,x}$ is uniquely defined in $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^d)\subset\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^d)$, satisfying

$$\|X^{t,x}\|^2_{\mathcal{S}^2}\le C(1+|x|^2),\tag{20}$$

$$\|X^{t,x}-X^{t',x'}\|^2_{\mathcal{S}^2}\le C(1+|x|^2)(|x-x'|^2+|t-t'|).\tag{21}$$

3.1 Existence, Uniqueness and Markovian estimates

In the next result, we show how Assumption 3.1 implies Assumption 2.1, so that Theorem 2.7 applies to the Markovian case.

Theorem 3.1. *Under Assumption 3.1, for any $(t,x)\in[0,T+\theta]\times\mathbb{R}^d$, the system* (19) *admits a unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^d)\times\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution.*

Proof. Let $\Phi^{t,x}=\Phi(\cdot,X^{t,x}_\cdot)$, $\Psi^{t,x}=\Psi(\cdot,X^{t,x}_\cdot)$ and, for $(\omega,s,y,z,\mu)\in\Omega\times[0,T]\times\mathbb{R}^n\times\mathbb{R}^{n\times q}\times\mathcal{P}_2(E)$, $f^{t,x}(\omega,s,y,z,\mu)=f(s,X^{t,x}_s(\omega),y,z,\mu)$. Note that $f^{t,x}$ is $Prog^t\otimes\mathcal{B}^n\otimes\mathcal{B}^{n\times q}\otimes\mathcal{B}(\mathcal{P}_2(E))$ measurable, where $Prog^t$ is the progressive sigma-algebra associated with $\mathfrak{F}^t$. Then the equations (19) for $(Y^{t,x},Z^{t,x})$ are of the form (2) with parameters $(f^{t,x},\Phi^{t,x},\Psi^{t,x},\mu)$. We need to prove that Assumption 2.1 holds (with filtration $\mathfrak{F}^t$) in order to apply Theorem 2.7.

Under Assumption 3.1(i), $\Phi^{t,x}$ and $\Psi^{t,x}$ are $\mathfrak{F}^t$ adapted and continuous, and (20) yields

$$\begin{aligned}\|\Phi^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)} &= \mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,X^{t,x}_s)|^2\right]\\ &\le 2\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,X^{t,x}_s)-\Phi(s,0)|^2\right]+2\mathbb{E}\left[\sup_{T\le s\le T+\theta}|\Phi(s,0)|^2\right]\\ &\le 2\Lambda^2\|X^{t,x}\|^2_{\mathcal{S}^2}+2\|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)}<\infty,\end{aligned}$$

as well as

$$\begin{aligned}\|\Psi^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} &= \mathbb{E}\left[\int_T^{T+\theta}|\Psi(s,X^{t,x}_s)|^2ds\right]\\ &\le 2\mathbb{E}\left[\int_T^{T+\theta}|\Psi(s,X^{t,x}_s)-\Psi(s,0)|^2ds\right]+2\mathbb{E}\left[\int_T^{T+\theta}|\Psi(s,0)|^2ds\right]\\ &\le 2\Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)}+2\|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)}<\infty.\end{aligned}$$

Hence Assumption 2.1(i) holds.

We have, for $(\omega,s,y,y',z,z',\mu,\mu')\in\Omega\times[0,T]\times(\mathbb{R}^n)^2\times(\mathbb{R}^{n\times q})^2\times(\mathcal{P}_2(E))^2$,

$$\begin{aligned}\left|f^{t,x}(\omega,s,y,z,\mu)-f^{t,x}(\omega,s,y',z',\mu')\right| &= \left|f(s,X^{t,x}_s(\omega),y,z,\mu)-f(s,X^{t,x}_s(\omega),y',z',\mu')\right|\\ &\le \Lambda\left(|y-y'|+|z-z'|+W_2(\mu,\mu')\right),\\ \mathbb{E}\left[\int_0^T|f^{t,x}(s,0,0,\mu^{0,0}_s)|^2ds\right] &= \mathbb{E}\left[\int_0^T|f(s,X^{t,x}_s,0,0,\mu^{0,0}_s)|^2ds\right]\\ &\le 2\Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T)}+2\|f(\cdot,0,0,0,\mu^{0,0})\|^2_{\mathcal{H}^2(T)}<\infty,\end{aligned}$$

so that Assumption 2.1(ii) holds, while Assumption 2.1(iii) holds by Assumption 3.1(iii). In conclusion, $(f^{t,x},\Phi^{t,x},\Psi^{t,x},\mu)$ satisfies Assumption 2.1. Theorem 2.7 thus ensures existence and uniqueness of an $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution $(Y^{t,x},Z^{t,x})$ to the Markovian ABSDE (19). □

Using the a priori estimates (9)-(10), we also obtain the following quadratic growth together with Hölder in time and Lipschitz in space dependence with respect to the initial condition.

Proposition 3.2. *Under Assumption 3.1, for $(t,x)\in[0,T+\theta]\times\mathbb{R}^d$, let $(X^{t,x},Y^{t,x},Z^{t,x})$ be the unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^d)\times\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution to (19) as per Theorem 3.1. Then there exists a constant $C\ge 0$ such that*

$$\|Y^{t,x}\|^2_{\mathcal{S}^2}+\|Z^{t,x}\|^2_{\mathcal{H}^2}\le C(1+|x|^2). \tag{22}$$

Moreover, for $0\le t,t'\le T$ and $x,x'\in\mathbb{R}^d$, there exists a constant $C'\ge 0$ such that

$$\|Y^{t,x}-Y^{t',x'}\|^2_{\mathcal{S}^2}+\|Z^{t,x}-Z^{t',x'}\|^2_{\mathcal{H}^2}\le C'(1+|x|^2)(|x-x'|^2+|t-t'|). \tag{23}$$

Proof. Note that $(Y,Z)=(0,0)$ is the unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution to (2) with parameters $(f,\Phi,\Psi,\mu)=(0,0,0,\mu)$.

From the a priori estimates (9)-(10), with β large enough, we obtain the existence of $K \geq 0$ such that

$$\begin{aligned}
&\|Y^{t,x}\|^2_{\mathcal{S}^2} + \|Z^{t,x}\|^2_{\mathcal{H}^2} \leq \|Y^{t,x}\|^2_{\mathcal{S}^2(T)} + \|Y^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)} + \|Z^{t,x}\|^2_{\mathcal{H}^2_\beta(T)} + \|Z^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} \\
&= \|Y^{t,x}\|^2_{\mathcal{S}^2(T)} + \|\Phi(\cdot, X^{t,x}_\cdot)\|^2_{\mathcal{S}^2(T,T+\theta)} + \|Z^{t,x}\|^2_{\mathcal{H}^2_\beta(T)} + \|\Psi(\cdot, X^{t,x}_\cdot)\|^2_{\mathcal{H}^2(T,T+\theta)} \\
&\leq C\left(\|f(\cdot, X^{t,x}_\cdot, 0, 0, \mu^{0,0}_\cdot)\|^2_{\mathcal{H}^2(T)} + \|\Phi(\cdot, X^{t,x}_\cdot)\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot, X^{t,x}_\cdot)\|^2_{\mathcal{H}^2(T,T+\theta)}\right).
\end{aligned}$$

We then compute, using Assumption 3.1,

$$\begin{aligned}
&\|Y^{t,x}\|^2_{\mathcal{S}^2} + \|Z^{t,x}\|^2_{\mathcal{H}^2} \\
&\leq 2C\left(\Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T)} + \|f(\cdot,0,0,0,\mu^{0,0}_\cdot)\|^2_{\mathcal{H}^2(T)} + \Lambda^2\|X^{t,x}\|^2_{\mathcal{S}^2(T,T+\theta)}\right. \\
&\qquad \left. + \|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)} + \Lambda^2\|X^{t,x}\|^2_{\mathcal{H}^2(T,T+\theta)} + \|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq 2C\left(\Lambda^2(T+1)\|X^{t,x}\|^2_{\mathcal{S}^2} + \|f(\cdot,0,0,0,\mu^{0,0}_\cdot)\|^2_{\mathcal{H}^2(T)}\right. \\
&\qquad \left. + \|\Phi(\cdot,0)\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot,0)\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C(1+|x|^2),
\end{aligned}$$

where we used (20) for the last inequality. This proves (22).

To derive (23), we compute, similarly to (9)-(10):

$$\begin{aligned}
&\|Y^{t,x} - Y^{t',x'}\|^2_{\mathcal{S}^2} + \|Z^{t,x} - Z^{t',x'}\|^2_{\mathcal{H}^2} \\
\leq&\|Y^{t,x} - Y^{t',x'}\|^2_{\mathcal{S}^2(T)} + \|\Phi(\cdot, X^{t,x}) - \Phi(\cdot, X^{t',x'})\|^2_{\mathcal{S}^2(T,T+\theta)} \\
&+ \|Z^{t,x} - Z^{t',x'}\|^2_{\mathcal{H}^2(T)} + \|\Psi(\cdot, X^{t,x}) - \Psi(\cdot, X^{t',x'})\|^2_{\mathcal{H}^2(T,T+\theta)} \\
\leq&C\left(\|f(\cdot, X^{t,x}, Y^{t',x'}, Z^{t',x'}, \mu^{Y^{t',x'}, Z^{t',x'}}) - f(\cdot, X^{t',x'}, Y^{t',x'}, Z^{t',x'}, \mu^{Y^{t',x'}, Z^{t',x'}})\|^2_{\mathcal{H}^2(T)}\right. \\
&\left. + \|\Phi(\cdot, X^{t,x}) - \Phi(\cdot, X^{t',x'})\|^2_{\mathcal{S}^2(T,T+\theta)} + \|\Psi(\cdot, X^{t,x}) - \Psi(\cdot, X^{t',x'})\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C\Lambda^2\left(\|X^{t,x} - X^{t',x'}\|^2_{\mathcal{H}^2(T)} + \|X^{t,x} - X^{t',x'}\|^2_{\mathcal{S}^2(T,T+\theta)} + \|X^{t,x} - X^{t',x'}\|^2_{\mathcal{H}^2(T,T+\theta)}\right) \\
&\leq C\Lambda^2(T+\theta+1)\|X^{t,x} - X^{t',x'}\|^2_{\mathcal{S}^2} \leq C'(1+|x|^2)(|x-x'|^2 + |t-t'|),
\end{aligned}$$

where we used (21) for the last inequality. □

3.2 Semigroup Properties

In this section, we prove that, in this Markovian setting, for each $0 \leq t \leq s \leq T+\theta$ and $x \in \mathbb{R}^d$, $(Y^{t,x}_s, Z^{t,x}_s)$ can be represented as a measurable function of time s and state $X^{t,x}_s$. This is similar to what is obtained for the classical BSDEs without anticipated feature in El Karoui et al. (1997), modulo the following result, which proves that the anticipated term is itself nothing but a ($\mathcal{P}_2(E)$ valued) function of time and state.

In order to deal with the anticipated term, we need the following additional assumption, which we prove to be satisfied in a fairly general setting.

Assumption 3.2. For any measurable $(u,v):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times q}$, there exists a measurable map $\mu^{u,v}:[0,T]\times\mathbb{R}^d\to\mathcal{P}_2(E)$ such that, for any $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$,

$$\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}=\mu^{u,v}(t,X_t^{t_0,x_0}),\quad a.s..$$

Lemma 3.3. *Assume that, for all measurable $(u,v):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times q}$, there exists a measurable map $A^{u,v}:[0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to E$ such that, for all $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$, $\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}=\mathbb{L}(A^{u,v}(t,X_t^{t_0,x_0},(W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)$ a.s.. Then μ satisfies Assumption 3.2, with $\mu^{u,v}$ defined by*

$$\mu^{u,v}(t,x)=\mathbb{L}(A^{u,v}(t,x,(W-W_t)\mathbf{1}_{[t,T+\theta]})),\ (t,x)\in[0,T]\times\mathbb{R}^d. \tag{24}$$

Proof. Fix a measurable map $(u,v):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times q}$, consider $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$ and $\mu^{u,v}$ as in (24). For any bounded continuous map $\varphi:E\to\mathbb{R}$, we have, almost surely,

$$\begin{aligned}\int_E\varphi(y)\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}(dy)&=\mathbb{E}\left[\varphi(A^{u,v}(t,X_t^{t_0,x_0},(W-W_t)\mathbf{1}_{[t,T+\theta]}))|\mathfrak{F}_t\right]\\&=\mathbb{E}\left[\varphi(A^{u,v}(t,x,(W-W_t)\mathbf{1}_{[t,T+\theta]}))\right]\big|_{x=X_t^{t_0,x_0}}\\&=\int_E\varphi(y)\mu^{u,v}(t,X_t^{t_0,x_0})(dy),\end{aligned}$$

where we used the fact that $X_t^{t_0,x_0}$ is $\mathfrak{F}_t$ measurable while $(W-W_t)\mathbf{1}_{[t,T+\theta]}$ is independent from $\mathfrak{F}_t$ for the second equality, and the definition of $\mu^{u,v}$ for the last equality. This proves that $\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}=\mu^{u,v}(t,X_t^{t_0,x_0})$ a.s., as desired. □

Proposition 3.4. *Assume that $\mu_t^{Y,Z}(\omega)=\mathbb{L}(A(t,(Y,Z,W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)(\omega)$ for almost all ω and for some map $A:[0,T]\times\mathcal{C}([0,T+\theta],\mathbb{R}^n)\times L^2([0,T+\theta],\mathbb{R}^{n\times q})\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to E$. Then μ satisfies Assumption 3.2, with $\mu^{u,v}$ defined by*

$$\mu^{u,v}(t,x)=\mathbb{L}(A(t,(u(\cdot,X_\cdot^{t,x}),v(\cdot,X_\cdot^{t,x}),(W_\cdot-W_t)\mathbf{1}_{[t,T+\theta]}))),\ (t,x)\in[0,T]\times\mathbb{R}^d.$$

Proof. First, X being a strong solution to the first equation in (19), by Touzi (2012, Section 2.2, p.10), there exists a measurable $\psi:[0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to\mathcal{C}([0,T+\theta],\mathbb{R}^d)$ such that, for any $(t,x)\in[0,T]\times\mathbb{R}^d$,

$$X_\cdot^{t,x}=\psi(t,x,W_\cdot-W_t). \tag{25}$$

By the flow property, $X^{t_0,x_0}\mathbf{1}_{[t,T+\theta]}$ is indistinguishable from $X^{t,X_t^{t_0,x_0}}\mathbf{1}_{[t,T+\theta]}$, hence, for any measurable $(u,v):[0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{n\times q}$, $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$, we almost surely have

$$\begin{aligned}&\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}\\&=\mathbb{L}(A(t,(u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0}),W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)\\&=\mathbb{L}(A(t,(u(\cdot,\psi(t,X_t^{t_0,x_0},W_\cdot-W_t)),v(\cdot,\psi(t,X_t^{t_0,x_0},W_\cdot-W_t)),W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t)\\&=\mathbb{L}(A^{u,v}(t,X_t^{t_0,x_0},(W-W_t)\mathbf{1}_{[t,T+\theta]})|\mathfrak{F}_t),\end{aligned}$$

with $A^{u,v}(t,x,w) := A(t,(u(\cdot,\psi(t,x,w)),v(\cdot,\psi(t,x,w)),w)\mathbf{1}_{[t,T+\theta]})$. The result then follows from the previous lemma. □

Proposition 3.5. *Assume that* $\mu_t^{Y,Z}(\omega) = \mathbb{L}(\int_t^{t+\theta} A(s,Y_s,Z_s)dW_s|\mathfrak{F}_t)(\omega)$ *for some* $\theta > 0$ *and some measurable map* $A : [0,T+\theta]\times\mathbb{R}^n\times\mathbb{R}^{n\times q} \to \mathbb{R}^{m\times q}$ *such that* $A(\cdot,Y,Z) \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^{m\times q})$ *for all* $(Y,Z) \in \mathcal{H}^2(\mathfrak{F},\mathbb{R}^n\times\mathbb{R}^{n\times q})$, *where* $m \geq 1$ *is an integer. Then* μ *satisfies Assumption 3.2, with* $\mu^{u,v}$ *defined by*

$$\mu^{u,v}(t,x) = \mathbb{L}(\int_t^{t+\theta} A(s,u(s,X_s^{t,x}),v(s,X_s^{t,x}))dW_s),\ (t,x)\in[0,T]\times\mathbb{R}^d.$$

Proof. For any measurable $(u,v) : [0,T+\theta]\times\mathbb{R}^d \to \mathbb{R}^n\times\mathbb{R}^{n\times q}$, $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$, we almost surely obtain using the same arguments and notations as in the previous proof,

$$\begin{aligned}
&\mu_t^{u(\cdot,X_\cdot^{t_0,x_0}),v(\cdot,X_\cdot^{t_0,x_0})}\\
&= \mathbb{L}(\int_t^{t+\theta} A(s,Y_s,Z_s)dW_s|\mathfrak{F}_t)\\
&= \mathbb{L}(\int_t^{t+\theta} A(s,u(s,X_s^{t_0,x_0}),v(s,X_s^{t_0,x_0}))dW_s|\mathfrak{F}_t)\\
&= \mathbb{L}(\int_t^{t+\theta} A(s,u(s,\psi(t,X_t^{t_0,x_0},W_\cdot-W_t)_s),v(s,\psi(t,X_t^{t_0,x_0},W_\cdot-W_t)_s))dW_s|\mathfrak{F}_t).
\end{aligned}$$

Observe that the inner random variable is $\sigma(X_t^{t_0,x_0},(W_s-W_t)_{s\in[t,t+\theta]})$ measurable, hence of the form $A^{u,v}(t,X_t^{t_0,x_0},(W_\cdot-W_t)\mathbf{1}_{[t,t+\theta]})$ for some measurable map $A^{u,v} : [0,T]\times\mathbb{R}^d\times\mathcal{C}([0,T+\theta],\mathbb{R}^q)\to\mathbb{R}^n$. The result then follows from the previous lemma. □

Remark 3.1. Following Remark 2.5 page 15, under Assumption 2.2, Assumption 3.2 reads: For any measurable $u : [0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n$, $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$, there exists a measurable map $\mu^u : [0,T]\times\mathbb{R}^d\to\mathcal{P}_2(E)$ such that, for any $(t_0,x_0)\in[0,T]\times\mathbb{R}^d$ and $t\in[t_0,T]$,

$$\mu_t^{u(\cdot,X_\cdot^{t_0,x_0})} = \mu^u(t,X_t^{t_0,x_0}),\quad \text{a.s..}$$

As in El Karoui et al. (1997) following Çinlar et al. (1980, Convention (3.6)-(ii) and Theorem (6.27)), let $\mathcal{E}$ denote the sigma algebra on $\mathbb{R}^d$ generated by the α excessive functions relative to W (see Blumenthal and Getoor (2007, Chapter II)). We then have the following representation result.

Theorem 3.6. *Under Assumptions 3.1 and 3.2, there exists a* $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ *measurable map*

$$(u,v) : [0,T+\theta]\times\mathbb{R}^d\to\mathbb{R}^n\times\mathbb{R}^{d\times n}$$

such that, for any $(t,x)\in[0,T]\times\mathbb{R}^d$, $u(t,x) = Y_t^{t,x}$ *and, ds* $\otimes$ $d\mathbb{P}$ *almost surely on* $[t,T+\theta]$,

$$Y_s^{t,x} = u(s,X_s^{t,x}) \quad \text{and} \quad Z_s^{t,x} = v(s,X_s^{t,x})^\top\sigma(s,X_s^{t,x}). \tag{26}$$

Moreover, the function u *is locally* $\frac{1}{2}$*-Hölder continuous in its first variable and locally Lipschitz continuous in its second variable.*

Proof. For any $(t,x) \in [0,T]\times\mathbb{R}^d$ and $(\widetilde{Y},\widetilde{Z}) \in \mathcal{H}^2(\mathfrak{F}^t, \mathbb{R}^n\times\mathbb{R}^{n\times q})$, we define $\Gamma^{t,x}(\widetilde{Y},\widetilde{Z}) = (Y,Z)$ as follows: $(Y,Z)_{|[0,T]}$ is defined as the unique $\mathcal{H}^2((\mathfrak{F}^t)^T, \mathbb{R}^n\times\mathbb{R}^{n\times q})$ solution to the classical BSDE

$$Y_s = \Phi(T, X_T^{t,x}) + \int_s^T f(r, X_r^{t,x}, Y_r, Z_r, \mu_r^{\widetilde{Y},\widetilde{Z}})dr - \int_s^T Z_r dW_r, \quad s\in[0,T],$$

and we set $(Y,Z) = (\Phi(\cdot, X_\cdot^{t,x}), \Psi(\cdot, X_\cdot^{t,x}))$ on $[T, T+\theta]$.

From Theorem 2.7(ii), we know that the sequence of processes $(Y^{t,x,k}, Z^{t,x,k})_{k\geq 0}$ defined by $(Y^{t,x,0}, Z^{t,x,0}) = (0,0)$ and, for $k\geq 0$, $(Y^{t,x,k+1}, Z^{t,x,k+1}) = \Gamma^{t,x}(Y^{t,x,k}, Z^{t,x,k})$, converges $ds\otimes d\mathbb{P}$ almost surely to $(Y^{t,x}, Z^{t,x})$.

We now prove by induction that, $ds\otimes d\mathbb{P}$ almost surely, for any $k\geq 0$, $Y_s^{t,x,k} = u_k(s, X_s^{t,x})$ and $Z_s^{t,x,k} = (v_k^\top\sigma)(s, X_s^{t,x})$ for some measurable map $(u_k, v_k) : ([0,T+\theta]\times\mathbb{R}^d, \mathcal{B}([0,T+\theta])\otimes\mathcal{E}) \to \mathbb{R}^n\times\mathbb{R}^{d\times n}$.

Since $(Y^{t,x,0}, Z^{t,x,0}) = (0,0)$, the property is satisfied for $k=0$. Assume that the property holds for some $k\geq 0$. From Assumption 3.2 applied to u_k and $(v_k^\top\sigma)$, we get $\mu_s^{Y^{t,x,k}, Z^{t,x,k}} = \mu^k(s, X_s^{t,x}) = \mu^{u_k, v_k^\top\sigma}(s, X_s^{t,x})$ where μ^k is a $\mathcal{P}_2(E)$ valued measurable map. Then, on $[t,T]$, $(Y^{t,x,k+1}, Z^{t,x,k+1})$ is the unique $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution to the classical BSDE

$$Y_s = \Phi_T + \int_s^T f^{k+1}(r, X_r^{t,x}, Y_r, Z_r)dr - \int_s^T Z_r dW_r,$$

where

$$f^{k+1} : [0,T]\times\mathbb{R}^d\times\mathbb{R}^n\times\mathbb{R}^{n\times q} \to \mathbb{R}^n$$
$$(t,x,y,z) \mapsto f(t,x,y,z,\mu^k(t,x)).$$

Applying El Karoui et al. (1997, Theorem 4.1), we obtain the existence of a $\mathcal{B}([0,T])\otimes\mathcal{E}$ measurable map $(u_{k+1}, v_{k+1}) : [0,T]\times\mathbb{R}^d \to \mathbb{R}^n\times\mathbb{R}^{d\times n}$ such that, for all $(t,x)\in[0,T]\times\mathbb{R}^d$, $Y_t^{t,x,k+1} = u_{k+1}(t,x)$ and $(Y_s^{t,x,k+1}, Z_s^{t,x,k+1}) = (u_{k+1}, v_{k+1}^\top\sigma)(s, X_s^{t,x})$, $ds\otimes d\mathbb{P}$ almost surely on $[t,T]$. We eventually set $(u_{k+1}, v_{k+1})(s,x) = (\Phi(s,x), \widetilde{\Psi}(s,x))$ for $(s,x)\in[T,T+\theta]\times\mathbb{R}^d$ (recall Assumption 3.1(i)), so $(Y^{t,x,k+1}, Z^{t,x,k+1}) = (u_{k+1}, v_{k+1}^\top\sigma)(\cdot, X^{t,x})$ holds $ds\otimes d\mathbb{P}$ almost surely on $[t,T+\theta]$.

We define measurable maps $u : [0,T+\theta]\times\mathbb{R}^d \to (\mathbb{R}\cup+\infty)^n$ and $z : [0,T+\theta]\times\mathbb{R}^d \to (\mathbb{R}\cup+\infty)^{n\times q}$ componentwise by

$$u^i(t,x) = \limsup_{k\to\infty} u_k^i(t,x), \qquad z^{i,j}(t,x) = \limsup_{k\to\infty}(v_k^\top\sigma)^{i,j}(t,x).$$

Since u_k and $v_k^\top\sigma$ are $\mathcal{B}([0,T+\theta])\otimes\mathcal{E}$ measurable, so are u and z. Moreover, by the $ds\otimes d\mathbb{P}$ a.s. convergence of $(Y^{t,x,k}, Z^{t,x,k})$ toward $(Y^{t,x}, Z^{t,x})$, we have, for any $(t,x)\in[0,T]\times\mathbb{R}^d$, $ds\otimes d\mathbb{P}$ almost surely on $[t,T+\theta]$,

$$u^i(s, X_s^{t,x}) = \limsup_{k\to\infty} u_k^i(s, X_s^{t,x}) = \lim_{k\to\infty} Y_s^{t,x,k,i} = Y_s^{t,x,i},$$
$$z^{i,j}(s, X_s^{t,x}) = \limsup_{k\to\infty}(v_k^\top\sigma)^{i,j}(s, X_s^{t,x}) = \lim_{k\to\infty} Z_s^{t,x,k,i,j} = Z_s^{t,x,i,j}.$$

Hence the u^i and the $z^{i,j}$ are in fact real valued and, $ds \otimes d\mathbb{P}$ almost surely on $[t, T+\theta]$,

$$u(s, X_s^{t,x}) = Y_s^{t,x} \quad \text{and} \quad z(s, X_s^{t,x}) = Z_s^{t,x}.$$

Now set $v = (\sigma^+)^\top z^\top$, where σ^+ denotes the Moore-Penrose pseudo-inverse of σ; this map is $\mathcal{B}([0, T+\theta]) \otimes \mathcal{E}$ measurable. Since, for every k,

$$(v_k^\top \sigma)(s, X_s^{t,x}) = (v_k^\top \sigma)(s, X_s^{t,x})\, \sigma(s, X_s^{t,x})^+ \sigma(s, X_s^{t,x}),$$

passing to the limit yields, $ds \otimes d\mathbb{P}$ almost surely on $[t, T+\theta]$,

$$v^\top \sigma(s, X_s^{t,x}) = z(s, X_s^{t,x})\, \sigma(s, X_s^{t,x})^+ \sigma(s, X_s^{t,x}) = z(s, X_s^{t,x}) = Z_s^{t,x}.$$

Finally, since $Y^{t,x,k} \to Y^{t,x}$ in $\mathcal{S}^2(\mathfrak{F}^t, \mathbb{R}^n)$ by Theorem 2.7(ii), and since $Y_t^{t,x,k}$ and $Y_t^{t,x}$ are $\mathfrak{F}_t^t$ measurable, hence deterministic, we have

$$|u_k(t,x) - Y_t^{t,x}|^2 = |Y_t^{t,x,k} - Y_t^{t,x}|^2 \le \|Y^{t,x,k} - Y^{t,x}\|_{\mathcal{S}^2}^2 \to 0.$$

Therefore $u(t,x) = \lim_{k\to\infty} u_k(t,x) = Y_t^{t,x}$.

Finally, let $(t, t', x, x') \in [0,T]^2 \times (\mathbb{R}^d)^2$, and assume e.g. $t \le t'$. By Proposition 3.2, there exists $C \ge 0$ such that

$$\begin{aligned} |u(t,x) - u(t,x')|^2 = |Y_t^{t,x} - Y_t^{t,x'}|^2 &\le \|Y^{t,x} - Y^{t,x'}\|_{\mathcal{S}^2}^2 \\ &\le C(1+|x|^2)|x-x'|^2. \end{aligned}$$

Thus $u(t,\cdot)$ is locally Lipschitz continuous, uniformly with respect to t. Moreover, using the BSDE between t and t', we obtain

$$u(t,x) - u(t',x) = \mathbb{E}\left[u(t', X_{t'}^{t,x}) - u(t',x) + \int_t^{t'} f(r, X_r^{t,x}, Y_r^{t,x}, Z_r^{t,x}, \mu_r^{Y^{t,x}, Z^{t,x}}) dr \right].$$

Therefore,

$$\begin{aligned} |u(t,x) - u(t',x)|^2 &\le 2\mathbb{E}\left[|u(t', X_{t'}^{t,x}) - u(t',x)|^2 \right] \\ &\quad + 2|t-t'|\mathbb{E}\left[\int_t^{t'} |f(r, X_r^{t,x}, Y_r^{t,x}, Z_r^{t,x}, \mu_r^{Y^{t,x}, Z^{t,x}})|^2 dr \right] \\ &\le C\mathbb{E}\left[|X_{t'}^{t,x} - x|^2 \right] + C|t-t'|(1+|x|^2) \\ &\le C(1+|x|^2)|t-t'|, \end{aligned}$$

which proves that $u(\cdot, x)$ is locally $\frac{1}{2}$-Hölder continuous. □

4 Feynman-Kac Representation

In the sequel, we denote by $\mathcal{A}$ the infinitesimal generator of X, namely, for smooth $\varphi : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ with row-gradient $\partial_x \varphi$ and Hessian matrix $\partial_{x^2}^2 \varphi$,

$$\mathcal{A}\varphi = \partial_x \varphi\, b + \frac{1}{2} \mathrm{Tr}[\sigma^\top (\partial_{x^2}^2 \varphi)\, \sigma].$$

The goal of this section is to link the values $(Y_t^{t,x}, Z_t^{t,x})$, $(t,x) \in [0,T]\times\mathbb{R}^d$ in Proposition 3.2 to a function $u : [0, T+\theta] \times \mathbb{R}^d \to \mathbb{R}^n$, solution to the PDE (with ∂_t also denoted ∂_1, for more clarity, in some probabilistic computations)

$$\begin{aligned} &u = \Phi \quad \text{on } [T, T+\theta] \times \mathbb{R}^d, \\ &\partial_t u + \mathcal{A}u + f(t, x, u, (\partial_x u)^\top \sigma, \mu^{u,z}) = 0 \quad \text{on } [0,T) \times \mathbb{R}^d, \end{aligned} \tag{27}$$

where $z = (\partial_x u)^\top \sigma \mathbf{1}_{[0,T]} + \Psi \mathbf{1}_{(T,T+\theta]}$ and $\mu^{u,z}$ is defined in Assumption 3.2.

Remark 4.1. Notice that the PDE (27), dubbed anticipated PDE hereafter, is nonlocal, not only in space (as usual for example for PDEs associated with jump-diffusions or with McKean-Vlasov or mean-field control PDEs, see e.g. Mou (2017) and the references therein), but also in time, due to the anticipated nature of the BSDE: the equation valued at (t,x) involves, in general, the values of u on $[t, T+\theta] \times \mathbb{R}^d$.

We start with a verification theorem, showing that a smooth solution to (27) provides a solution to the Markovian ABSDE (19).

Lemma 4.1 (Markovian ABSDE verification theorem)**.** *Under Assumption 3.2, suppose that the nonlocal PDE* (27) *admits a regular solution* $u \in \mathcal{C}^{1,2}([0, T+\theta] \times \mathbb{R}^d; \mathbb{R}^n)$ *such that, for any* $(t,x) \in [0, T+\theta] \times \mathbb{R}^d$,

$$|u(t,x)| + |((\partial_x u)^\top \sigma)(t,x)| \le C(1+|x|).$$

Then, under Assumption 3.1, for each $(t,x) \in [0,T] \times \mathbb{R}^d$, *we have* $(Y^{t,x}, Z^{t,x}) = (u, (\partial_x u)^\top \sigma \mathbf{1}_{[0,T]} + \Psi \mathbf{1}_{(T,T+\theta]})(\cdot, X^{t,x})$ *on* $[t, T+\theta]$.

Proof. Letting $(Y_s, Z_s) = (u, z)(s, X_s^{t,x})$ for $s \in [t, T+\theta]$, where we set $z = (\partial_x u)^\top \sigma \mathbf{1}_{[0,T]} + \Psi \mathbf{1}_{(T,T+\theta]}$, we first observe, by (27), that $Y_s = \Phi(s, X_s^{t,x})$ and $Z_s = \Psi(s, X_s^{t,x})$, $s \in (T, T+\theta]$.

For $s \in [t,T]$, applying Itô's formula to the smooth function u and the fact that u solves (27), we obtain

$$\begin{aligned} \Phi(T, X_T^{t,x}) &= Y_s + \int_s^T (\partial_1 + \mathcal{A}) u(r, X_r^{t,x}) dr + \int_s^T ((\partial_x u)^\top \sigma)(r, X_r^{t,x}) dW_r \\ &= Y_s - \int_s^T f(\cdot, u, ((\partial_x u)^\top \sigma), \mu^{u,z})(r, X_r^{t,x}) dr + \int_s^T ((\partial_x u)^\top \sigma)(r, X_r^{t,x}) dW_r \\ &= Y_s - \int_s^T f(r, X_r^{t,x}, Y_r, Z_r, \mu^{u,z}) dr + \int_s^T ((\partial_x u)^\top \sigma)(r, X_r^{t,x}) dW_r. \end{aligned}$$

From Assumption 3.2, we obtain, for any $r \in [t,T]$,

$$\mu^{u,z}(r, X_r^{t,x}) = \mu_r^{u(\cdot, X_\cdot^{t,x}), z(\cdot, X_\cdot^{t,x})} = \mu_r^{Y,Z}, \quad a.s.,$$

hence

$$\Phi(T, X_T^{t,x}) = Y_s - \int_s^T f(r, X_r^{t,x}, Y_r, Z_r, \mu_r^{Y,Z}) dr + \int_s^T ((\partial_x u)^\top \sigma)(r, X_r^{t,x}) dW_r.$$

This proves that (Y,Z) is a $\mathcal{S}^2(\mathfrak{F}^t,\mathbb{R}^n)\times\mathcal{H}^2(\mathfrak{F}^t,\mathbb{R}^{n\times q})$ solution to (19), hence $(Y,Z)=(Y^{t,x},Z^{t,x})$, by Theorem 3.1. □

Conversely, when $n=1$, using the comparison theorem 2.8, we prove that the value function u of a Markovian ABSDE furnishes a viscosity solution to the PDE (27). Toward this aim, we have to work under Assumption 2.2 guaranteeing that the anticipated term does not depend on the Z coordinate of the ABSDE solution. In addition, in our Markovian context, the monotonicity Assumption 2.3 takes the following form.

Assumption 4.1 (Markovian monotonicity)**.** Under Assumption 2.2, for any $(t,x)\in[0,T]\times\mathbb{R}^d$ and $(\widetilde{Y},\widetilde{Y}')\in\mathcal{H}^2(\mathfrak{F},\mathbb{R})^2$, if $\widetilde{Y}\geq\widetilde{Y}'\ dt\otimes\mathbb{P}$ a.e., then, for each $(s,y,z)\in[0,T]\times\mathbb{R}\times\mathbb{R}^q$,

$$f(s,X_s^{t,x}(\omega),y,z,\widetilde{\mu}_s^{\widetilde{Y}}(\omega))\geq f(s,X_s^{t,x}(\omega),y,z,\widetilde{\mu}_s^{\widetilde{Y}'}(\omega)),\text{ a.e..}$$

Last, we also need the following:

Assumption 4.2 (Time-continuity of the anticipated term)**.** For each $Y\in\mathcal{H}^2(\mathfrak{F},\mathbb{R})$,

$$\lim_{t\to 0}\mathbb{E}\left[W_2(\mu_t^Y,\mu_0^Y)^2\right]=0.$$

Definition 4.3 (Viscosity solution to the anticipated PDE)**.** Let $u\in\mathcal{C}([0,T+\theta]\times\mathbb{R}^d)$. We say that u is a viscosity subsolution (resp. supersolution) to (27) if $u=\Phi$ on $[T,T+\theta]\times\mathbb{R}^d$ and, for each $(t,x)\in[0,T)\times\mathbb{R}^d$ and $\phi\in\mathcal{C}^\infty([0,T+\theta]\times\mathbb{R}^d)$ such that $\min\phi-u=\phi(t,x)-u(t,x)=0$ (resp. $\max\phi-u=\phi(t,x)-u(t,x)=0$), we have

$$\partial_t\phi(t,x)+\mathcal{A}\phi(t,x)+f(t,x,\phi(t,x),((\partial_x\phi)^\top\sigma)(t,x),\mu^\phi(t,x))\geq 0$$
$$(\text{resp. }\partial_t\phi(t,x)+\mathcal{A}\phi(t,x)+f(t,x,\phi(t,x),((\partial_x\phi)^\top\sigma)(t,x),\mu^\phi(t,x))\leq 0).$$

Remark 4.2. Note that the above definition uses global test functions, i.e. ϕ achieves a global minimum (resp. maximum) of $\phi-u$ at (t,x), which is standard for such nonlocal equations (Mou, 2017).

Theorem 4.2 (The ABSDE value function is a viscosity solution to the anticipated PDE)**.** *Under Assumption 2.2, let (f,Φ,μ) be standard parameters satisfying Assumptions 3.1, 4.1 and 4.2. We consider the map $u(t,x)=Y_t^{t,x}$, as per Theorem 3.6. Then u is a viscosity solution to* (27).

Proof. For $(t,x)\in[T,T+\theta]\times\mathbb{R}^d$, one has $u(t,x)=Y_t^{t,x}=\Phi(t,x)$, so the first line of (27) is satisfied at (t,x). We now prove that u is a subsolution of the second line of (27). The proof that u is a supersolution of the second line of (27) is similar and left to the reader.

For $\mathrm{x}\in\mathbb{R}^d$, we consider a smooth test function $\phi\in\mathcal{C}^\infty([0,T+\theta]\times\mathbb{R}^d,\mathbb{R})$ with bounded derivatives at all order and such that $\min_{[0,T+\theta]\times\mathbb{R}^d}(\phi-u)=(\phi-u)(0,\mathrm{x})=0$ (note that having set time to 0 in $(0,\mathrm{x})$ is without loss of generality). We need to show that $g(0)\geq 0$, where

$$g(t)=\partial_t\phi(t,\mathrm{x})+\mathcal{A}\phi(t,\mathrm{x})+f(t,\mathrm{x},\phi(t,\mathrm{x}),\left((\partial_x\phi)^\top\sigma\right)(t,\mathrm{x}),\mu^\phi(t,\mathrm{x})),\ t\in[0,T]$$

and μ^ϕ comes from Assumption 3.2, see also Remark 3.1. For later use, we remark that $t \mapsto \mu^\phi(t,\mathrm{x})$, hence g, is continuous at 0. Indeed, we compute by Assumption 2.1(iii), for $t \in [0,T]$,

$$
\begin{aligned}
&W_2(\mu^\phi(t,\mathrm{x}),\mu^\phi(0,\mathrm{x}))^2 = W_2(\mu_t^{\phi(\cdot,X_\cdot^{t,\mathrm{x}})},\mu_0^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2 = \mathbb{E}\left[W_2(\mu_t^{\phi(\cdot,X_\cdot^{t,\mathrm{x}})},\mu_0^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2\right] \\
\le& 2\mathbb{E}\left[W_2(\mu_t^{\phi(\cdot,X_\cdot^{t,\mathrm{x}})},\mu_t^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2\right] + 2\mathbb{E}\left[W_2(\mu_t^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})},\mu_0^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2\right] \\
\le& 2\mathbb{E}\left[\int_t^{T+\theta} |\phi(u,X_u^{t,\mathrm{x}}) - \phi(u,X_u^{0,\mathrm{x}})|^2 du\right] + 2\mathbb{E}\left[W_2(\mu_t^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})},\mu_0^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2\right] \\
\le& C(1+|x|^2)|t-0| + 2\mathbb{E}\left[W_2(\mu_t^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})},\mu_0^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})})^2\right].
\end{aligned}
$$

where we used (21), and the continuity hypothesis in Assumption 4.2 ensures that the second term goes to 0 as $t \to 0$.

First, for any $0 < h < T$, one easily observes that $(Y^{0,\mathrm{x}}, Z^{0,\mathrm{x}})$ is the unique $\mathcal{S}^2(\mathfrak{F},\mathbb{R}) \times \mathcal{H}^2(\mathfrak{F},\mathbb{R}^q)$ solution on $[0, T+\theta]$ to the ABSDE with maturity h, terminal delay $T+\theta-h$ and standard parameters $(f^{0,\mathrm{x}}, u(\cdot, X_\cdot^{0,\mathrm{x}}) = Y^{0,\mathrm{x}}, \mu)$, with $f^{0,\mathrm{x}}(\omega,t,y,z,\mu) = f(t, X_t^{0,\mathrm{x}}(\omega), y, z, \mu)$ for any $(\omega,t,y,z,\mu) \in \Omega \times [0,h] \times \mathbb{R} \times \mathbb{R}^q \times \mathcal{P}_2(E)$, i.e., for $t \in [0, h+(T+\theta)-h] = [0, T+\theta]$,

$$
Y_t^{0,\mathrm{x}} = u\big(t \vee h, X_{t\vee h}^{0,\mathrm{x}}\big) + \int_{t\wedge h}^h f(s, X_s^{0,\mathrm{x}}, Y_s^{0,\mathrm{x}}, Z_s^{0,\mathrm{x}}, \mu_s^{Y^{0,\mathrm{x}}})ds - \int_{t\wedge h}^h Z_s^{0,\mathrm{x}} dW_s.
$$

Second, for fixed $0 < h < T$, we denote by $(\bar{Y}^h, \bar{Z}^h)$ the unique $\mathcal{S}^2(\mathfrak{F},\mathbb{R}) \times \mathcal{H}^2(\mathfrak{F},\mathbb{R}^q)$ solution on $[0, T+\theta]$ of the ABSDE with maturity h, terminal delay $T+\theta-h$ and standard parameters $(f^{0,\mathrm{x}}, \phi(\cdot, X^{0,\mathrm{x}}), \mu)$, i.e., for $t \in [0, T+\theta]$,

$$
\bar{Y}_t^h = \phi\big(t \vee h, X_{t\vee h}^{0,\mathrm{x}}\big) + \int_{t\wedge h}^h f(s, X_s^{0,\mathrm{x}}, \bar{Y}_s^h, \bar{Z}_s^h, \mu^{\bar{Y}^h})ds - \int_{t\wedge h}^h \bar{Z}_s^h dW_s. \tag{28}
$$

The inequality $\phi(\cdot, X_\cdot^{0,\mathrm{x}}) \ge u(\cdot, X_\cdot^{0,\mathrm{x}})$ holds on $[h, T+\theta]$, Assumption 2.2 is in force and $(f^{0,\mathrm{x}}, \mu)$ satisfies Assumption 2.3 (by Assumption 4.1). Hence the comparison theorem 2.8 yields $\bar{Y}_0^h \ge Y_0^{0,\mathrm{x}} = u(0,\mathrm{x}) = \phi(0,\mathrm{x})$.

We define, for $0 < h < T$ and $t \in [0, T+\theta]$, $\widetilde{Y}_t^h = \bar{Y}_t^h - \phi(t, X_t^{0,\mathrm{x}}) - \int_{t\wedge h}^h g(s)ds$, so that $\widetilde{Y}_t^h = 0$ for $t \ge h$. Since $\bar{Y}_0^h \ge \phi(0,\mathrm{x})$, we have that $\widetilde{Y}_0^h \ge -\int_0^h g(s)ds$. Assuming in addition that $h\epsilon(h) \ge \widetilde{Y}_0^h$ for a map ϵ defined in a neighborhood of 0 and such that $\epsilon(h) \to_{h\to 0} 0$, we obtain $\frac{1}{h}\int_0^h g(s)ds \ge -\epsilon(h)$, giving in the limit $h \to 0$ the required inequality $g(0) \ge 0$, by continuity of g at time 0.

To conclude the proof, we now prove that there indeed exists such a map ϵ.

By (28) and Itô's lemma, we observe that, for $t \in [0,h]$,

$$\begin{aligned}\widetilde{Y}_t^h =&\phi(h, X_h^{0,\mathrm{x}}) + \int_t^h f(s, X_s^{0,\mathrm{x}}, \bar{Y}_s^h, \bar{Z}_s^h, \mu_s^{\bar{Y}^h})ds - \int_t^h \bar{Z}_s^h dW_s \\ &- \phi(h, X_h^{0,\mathrm{x}}) + \int_t^h (\partial_1 + \mathcal{A})\phi(s, X_s^{0,\mathrm{x}})ds \\ &+ \int_t^h ((\partial_x \phi)^\top \sigma)(s, X_s^{0,\mathrm{x}})dW_s - \int_t^h g(s)ds \\ =&\int_t^h \gamma^h(s, X_s^{0,\mathrm{x}}, \widetilde{Y}_s^h, \widetilde{Z}_s^h, \mu_s^{h,\widetilde{Y}^h})ds - \int_t^h \widetilde{Z}_s^h dW_s,\end{aligned}$$

where we set

$$\begin{aligned}&\widetilde{Z}_s^h = \bar{Z}_s^h - ((\partial_x \phi)^\top \sigma)(s, X_s^{0,\mathrm{x}}), \\ &\gamma^h(s,x,y,z,\mu) = f\left(s, x, y + \phi(s,x) + \int_s^h g(r)dr, z + ((\partial_x \phi)^\top \sigma)(s,x), \mu\right) \\ &\qquad\qquad + (\partial_1 + \mathcal{A})\phi(s,x) - g(s), \text{ and} \\ &\mu^h : Y \mapsto \mu^{Y + \phi(\cdot, X^{0,\mathrm{x}}) + \int_{\cdot\wedge h}^h g(r)dr}.\end{aligned}$$

meaning that $(\widetilde{Y}^h, \widetilde{Z}^h)$ is the unique $\mathcal{S}^2(\mathfrak{F}, \mathbb{R}) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^q)$ solution to the ABSDE with maturity h, terminal delay $T + \theta - h$ and standard parameters $(\gamma^h, 0, \mu^h)$.

We thus have

$$\begin{aligned}\widetilde{Y}_0^h =&\mathbb{E}\left[\int_0^h \gamma^h(s, X_s^{0,\mathrm{x}}, \widetilde{Y}_s^h, \widetilde{Z}_s^h, \mu_s^{h,\widetilde{Y}^h})ds\right] \\ =&\mathbb{E}\left[\int_0^h \gamma^h(s, X_s^{0,\mathrm{x}}, \widetilde{Y}_s^h, \widetilde{Z}_s^h, \mu_s^{h,\widetilde{Y}^h}) - \gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})ds\right] \\ &+ \mathbb{E}\left[\int_0^h \gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})ds\right].\end{aligned} \tag{29}$$

From the definition of γ^h, we obtain, using Assumption 3.1(iii), the Cauchy-Schwarz inequality, Assumption 3.2 (remembering that $\widetilde{Y}^h = 0$ on $[h, T+\theta]$), and eventually

Proposition 2.4 (recalling that $(0,0)$ is solution of the ABSDE with parameters $(0,0,0)$):

$$
\begin{aligned}
&\mathbb{E}\left[\int_0^h \gamma^h(s, X_s^{0,\mathrm{x}}, \widetilde{Y}_s^h, \widetilde{Z}_s^h, \mu_s^{h,\widetilde{Y}^h}) - \gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0}) ds\right] \\
&\leq \Lambda \mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h| + |\widetilde{Z}_s^h| + W_2(\mu_s^{h,\widetilde{Y}^h}, \mu_s^{h,0}) ds\right] \\
&\leq \Lambda\sqrt{h}\left(\mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h |\widetilde{Z}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h W_2(\mu_s^{h,\widetilde{Y}^h}, \mu_s^{h,0})^2 ds\right]^{\frac{1}{2}}\right) \\
&\leq \Lambda\sqrt{h}\left(\mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h |\widetilde{Z}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h \mathbb{E}_s\left[\int_s^h |\widetilde{Y}_r^h|^2 dr\right] ds\right]^{\frac{1}{2}}\right) \\
&\leq \Lambda\sqrt{h}\left(\mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h |\widetilde{Z}_s^h|^2 ds\right]^{\frac{1}{2}} + \sqrt{h}\mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h|^2 ds\right]^{\frac{1}{2}}\right) \\
&\leq 2\Lambda\sqrt{h}\left(\mathbb{E}\left[\int_0^h |\widetilde{Y}_s^h|^2 ds\right]^{\frac{1}{2}} + \mathbb{E}\left[\int_0^h |\widetilde{Z}_s^h|^2 ds\right]^{\frac{1}{2}}\right) \\
&\leq C\sqrt{h}\mathbb{E}\left[\int_0^h |\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2 ds\right]^{\frac{1}{2}},
\end{aligned}
$$

where the constant K, which comes from the a priori estimates of Proposition 2.4, is independent of h. Indeed, in (7), the time horizon appears only through the ratio $\frac{T}{\nu^2}$. For the above ABSDE, this term is replaced by $\frac{h}{\nu^2}$, and $h \leq T$. Therefore, any choice $(\beta, \nu, \lambda, \eta)$ satisfying (7) for the horizon T also satisfies the corresponding condition for any horizon $h \leq T$, yielding a uniform constant $K = K(T, \theta, \Lambda)$.

We similarly obtain by the Cauchy-Schwarz inequality:

$$
\mathbb{E}\left[\int_0^h \gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0}) ds\right] \leq \sqrt{h}\mathbb{E}\left[\int_0^h |\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2 ds\right]^{\frac{1}{2}}.
$$

Plugging these estimates in (29), we obtain

$$
\widetilde{Y}_0^h \leq (K+1)\sqrt{h}\mathbb{E}\left[\int_0^h |\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2 ds\right]^{\frac{1}{2}} \leq h\epsilon(h),
$$

with $\epsilon(h) = (K+1)\sup_{0\leq s\leq h}\mathbb{E}\left[|\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2\right]^{\frac{1}{2}}$. It remains to prove that

$\epsilon(h) \underset{h\to 0}{\longrightarrow} 0$. Toward this aim, for $0 \le s \le h$, we compute by Assumption 3.1(iii):

$$
\begin{aligned}
&\mathbb{E}\left[|\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2\right] \\
=&\mathbb{E}\left[|f\left(s, X_s^{0,\mathrm{x}}, \phi(s, X_s^{0,\mathrm{x}}) + \int_s^h g(r)dr, ((\partial_x\phi)^\top\sigma)(s, X_s^{0,\mathrm{x}}), \mu_s^{h,0}\right)\right. \\
&\left. +(\partial_1 + \mathcal{A})\phi(s, X_s^{0,\mathrm{x}}) - (\partial_1 + \mathcal{A})\phi(s,\mathrm{x}) - f\left(s, \mathrm{x}, \phi(s,\mathrm{x}), ((\partial_x\phi)^\top\sigma)(s,\mathrm{x}), \mu^\phi(s,\mathrm{x})\right)|^2\right] \\
\le& C\mathbb{E}\left[|X_s^{0,\mathrm{x}} - x|^2\right] + C\mathbb{E}\left[|\phi(s, X_s^{0,\mathrm{x}}) - \phi(s,\mathrm{x})|^2\right] + C\mathbb{E}\left[|\int_s^h g(r)dr|^2\right] \\
&+ C\underbrace{\mathbb{E}\left[|((\partial_x\phi)^\top\sigma)(s, X_s^{0,\mathrm{x}}) - ((\partial_x\phi)^\top\sigma)(s,\mathrm{x})|^2\right]}_{I} + C\underbrace{\mathbb{E}\left[W_2(\mu_s^{h,0}, \mu^\phi(s,\mathrm{x}))^2\right]}_{II} \\
&+ 2\underbrace{\mathbb{E}\left[|(\partial_1 + \mathcal{A})\phi(s, X_s^{0,\mathrm{x}}) - (\partial_1 + \mathcal{A})\phi(s,\mathrm{x})|^2\right]}_{III}.
\end{aligned}
$$

Now, since ϕ is Lipschitz-continuous in the space variable, from Assumption 3.1(iv), we have

$$
\begin{aligned}
I &\le 2\mathbb{E}\left[|\partial_x\phi(s, X_s^{0,\mathrm{x}})|^2|\sigma(s, X_s^{0,\mathrm{x}}) - \sigma(s,\mathrm{x})|^2\right] + 2\mathbb{E}\left[|\partial_x\phi(s, X_s^{0,\mathrm{x}}) - \partial_x\phi(s,\mathrm{x})|^2|\sigma(s,\mathrm{x})|^2\right] \\
&\le C(1+|x|)^2\mathbb{E}\left[|X_s^{0,\mathrm{x}} - x|^2\right].
\end{aligned}
$$

A similar computation yields $III \le C(1+|x|)^2\mathbb{E}\left[|X_s^{0,\mathrm{x}} - x|^2\right]$. Assumption 3.1(iii) imply (as $0 \le s \le h$)

$$
\begin{aligned}
II &= \mathbb{E}\left[W_2(\mu_s^{h,0}, \mu_s^{\phi(\cdot,X^{0,\mathrm{x}})})^2\right] = \mathbb{E}\left[W_2(\mu_s^{\phi(\cdot,X_\cdot^{0,\mathrm{x}})+\int_{\cdot\wedge h}^h g(r)dr}, \mu_s^{\phi(\cdot,X^{0,\mathrm{x}})})^2\right] \\
&\le \mathbb{E}\left[\int_s^{T+\theta}\left|\int_{r\wedge h}^h g(\zeta)d\zeta\right|^2 dr\right] \le \mathbb{E}\left[\int_0^h\left|\int_r^h g(\zeta)d\zeta\right|^2 ds\right],
\end{aligned}
$$

as the inner integral obviously vanishes for $r \ge h$. Hence, from Jensen's inequality,

$$
\begin{aligned}
II &\le \mathbb{E}\left[\int_0^h (h-r)\int_r^h g(\zeta)^2 d\zeta dr\right] \\
&\le h\mathbb{E}\left[\int_0^h\int_0^h g(\zeta)^2 d\zeta dr\right] \\
&= h^2\mathbb{E}\left[\int_0^h g(\zeta)^2 d\zeta\right].
\end{aligned}
$$

Wrapping up,

$$
\begin{aligned}
\mathbb{E}\left[|\gamma^h(s, X_s^{0,\mathrm{x}}, 0, 0, \mu_s^{h,0})|^2\right] &\le C\mathbb{E}\left[|X_s^{0,\mathrm{x}} - \mathrm{x}|^2\right] + Ch\mathbb{E}\left[\int_0^h |g(r)|^2 dr\right] \\
&\le C\sup_{0\le r\le h}\mathbb{E}\left[|X_r^{0,\mathrm{x}} - \mathrm{x}|^2\right] + Ch\mathbb{E}\left[\int_0^h |g(r)|^2 dr\right],
\end{aligned}
$$

which goes to 0 as h goes to 0, by continuity of g at time 0, and by classical SDE estimates. □

5 An Approximation Scheme

As an application of the above results, we study a time-discretization scheme for the Markovian ABSDE (19), in line with the one for classical BSDEs originally introduced in Bouchard and Touzi (2004). In order to obtain a convergence result, we need to make the following additional assumptions on the data.

Assumption 5.1. Φ, Ψ, f and M are Hölder-continuous in time, uniformly in space:
(i) For all $(t,s,x) \in [T, T+\theta]^2 \times \mathbb{R}^d$, $|\Phi(t,x) - \Phi(s,x)| + |\Psi(t,x) - \Psi(s,x)| \leq \Lambda |t-s|^{\frac{1}{2}}$,
(ii) For any $(t,s,x,y,z,\mu) \in [0,T]^2 \times \mathbb{R}^d \times \mathbb{R} \times \mathbb{R}^q \times \mathcal{P}_2(E)$,

$$|f(t,x,y,z,\mu) - f(s,x,y,z,\mu)| \leq \Lambda |t-s|^{\frac{1}{2}},$$

(iii) For any $(t,s,Y,Z) \in [0,T]^2 \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}) \times \mathcal{H}^2(\mathfrak{F}, \mathbb{R}^q)$

$$W_2(M_t^{Y,Z}, M_s^{Y,Z}) \leq \Lambda |t-s|^{\frac{1}{2}}.$$

For x fixed in $\mathbb{R}^d$, we consider the solution (X,Y,Z) to (19) with initial condition $(0, \mathrm{x})$ of X.

Given integers $N' \geq N \geq 1$, we fix a time-grid $\{0 = t_0 < \cdots < t_N = T < t_{N+1} < \cdots < t_{N'} = T + \theta\}$ of mesh size $\mathrm{h} = \max_{0 \leq n < N'} \mathrm{h}_n$, where $\mathrm{h}_n = t_{n+1} - t_n$.

Defining

$$\widetilde{Z}^{\mathrm{h}}_{t_n} = \mathrm{h}_n^{-1} \mathbb{E}_{t_n} \left[\int_{t_n}^{t_{n+1}} Z_s ds \right], \quad 0 \leq n < N', \tag{30}$$

we have the following result (Zhang (2001)).

Lemma 5.1. *Under Assumptions 3.1 and 5.1, there exists a constant $C \geq 0$ such that,*

$$\sum_{n=0}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] dt \leq C \mathrm{h}.$$

Proof. In the Markovian setting, from Assumption 3.2 and Theorem 3.6, the anticipated term $M_t^{Y,Z}$ can be written as $\widetilde{M}(t, X_t)$, where $\widetilde{M} = M^{u, v^\top \sigma}$ is a deterministic ($\mathcal{P}_2(E)$ valued) map. Hence, (Y,Z) is the solution to the classical BSDE

$$Y_t = \Phi(T, X_T) + \int_t^T f(s, X_s, Y_s, Z_s, \widetilde{M}(s, X_s)) ds - \int_t^T Z_s dW_s, \quad 0 \leq t \leq T,$$

Applying Zhang (2001, Theorem 3.4.3), we have

$$\sum_{n=0}^{N-1} \int_{t_n}^{t_{n+1}} \mathbb{E} \left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2 \right] dt \leq C \mathrm{h}.$$

It remains to control the sum over $N \le n < N'$, which is easily done as follows, since $Z = \Psi(\cdot, X)$ on $[T, T+\theta]$. By Jensen's inequality and using Assumptions 3.1 and 5.1(i), we have

$$\begin{aligned}
&\sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_t - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] dt \\
\le& 2\sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_t - Z_{t_n}|^2\right] dt + 2\sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E}\left[|Z_{t_n} - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] \\
\le& 2\sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|\Psi(t, X_t) - \Psi(t_n, X_{t_n})|^2\right] dt \\
&+ 2\sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E}\left[|\Psi(t_n, X_{t_n}) - \mathrm{h}_n^{-1}\mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \Psi(s, X_s) ds\right]|^2\right] \\
\le& C\mathrm{h} + 2\sum_{n=N}^{N'-1} \mathrm{h}_n \mathbb{E}\left[|\mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \Psi(t_n, X_{t_n}) - \Psi(s, X_s)\frac{ds}{\mathrm{h}_n}\right]|^2\right] \\
&\le C\mathrm{h} + 2\sum_{n=N}^{N'-1} \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|\Psi(t_n, X_{t_n}) - \Psi(s, X_s)|^2\right] ds \\
&\le C\mathrm{h},
\end{aligned}$$

where $C \ge 0$ is a constant that can change from line to line (but is independent of h). □

In order to discretise in time the Markovian ABSDE, we first consider X^{h} defined by the (continuous-time version of the) Euler-Maruyama scheme X^{h} of X, defined by $X^{\mathrm{h}}_{t_0} = \mathrm{x}$ and, for $t_n \le t \le t_{n+1}, 0 \le n < N'$,

$$X^{\mathrm{h}}_t = X^{\mathrm{h}}_{t_n} + b(t_n, X^{\mathrm{h}}_{t_n})(t - t_n) + \sigma(t_n, X^{\mathrm{h}}_{t_n})(W_t - W_{t_n}).$$

By Kloeden and Platen (1992, Theorem 10.2.2),

$$\mathbb{E}\left[\sup_{0\le t\le T+\theta} |X^{\mathrm{h}}_t - X_t|^2\right] \le C\mathrm{h}. \tag{31}$$

Under Assumption 3.1(i), we readily obtain the following.

Lemma 5.2. *Under Assumption 3.1(i), we have*

$$\mathbb{E}\left[\sup_{T\le t\le T+\theta} |\Phi(t, X^{\mathrm{h}}_t) - \Phi(t, X_t)|^2\right] + \mathbb{E}\left[\sup_{T<t\le T+\theta} |\Psi(t, X^{\mathrm{h}}_t) - \Psi(t, X_t)|^2\right] \le \Lambda^2\mathrm{h}.$$

Regarding the ABSDE, we introduce the following scheme: $Y^{\mathrm{h}}_{t_n} = \Phi(t_n, X^{\mathrm{h}}_{t_n})$ and $Z^{\mathrm{h}}_{t_n} = \Psi(t_n, X^{\mathrm{h}}_{t_n})$, $N \le n \le N'$, and, for $0 \le n < N$,

$$\begin{aligned}
Z^{\mathrm{h}}_{t_n} &= \mathrm{h}_n^{-1}\mathbb{E}_{t_n}\left[(W_{t_{n+1}} - W_{t_n})Y^{\mathrm{h}}_{t_{n+1}}\right], \\
Y^{\mathrm{h}}_{t_n} &= \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h}}_{t_n})\mathrm{h}_n,
\end{aligned}$$

where, for $0 \leq n < N$, $M^{\mathrm{h}}_{t_n}$ is a $\mathcal{P}_2(E)$ valued, $\mathfrak{F}_{t_n}$ measurable random variable satisfying to

$$W_2^2(M^{\mathrm{h}}_{t_n}, M^{Y,Z}_{t_n}) \leq \mathbb{E}_{t_n}\left[\sum_{k=n}^{N'-1}\int_{t_k}^{t_{k+1}} |Y^{\mathrm{h}}_{t_k} - Y_s|^2 + |Z^{\mathrm{h}}_{t_k} - Z_s|^2 + |W_{t_k} - W_s|^2 ds\right]. \quad (32)$$

Remark 5.1. Equation (32) is satisfied under Assumption 3.1(iii) by taking

$$M^{\mathrm{h}}_{t_n} = M^{\sum_{m=n}^{N'-1}(Y^{\mathrm{h}}_{t_m}, Z^{\mathrm{h}}_{t_m})\mathbf{1}_{(t_m, t_{m+1}]}}. \quad (33)$$

The inequality (32) is more general and accounts for more general anticipating terms, depending also on the Brownian motion W, which is only discretly observed within the scheme, leading to additional approximation errors of the form (32).

Lemma 5.3 (The scheme in discrete time is well-defined)**.** *Under Assumption 3.1, $Y^{\mathrm{h}}_{t_n}$ is uniquely defined in $L^2(\mathfrak{F}_{t_n}, \mathbb{R})$ as soon as $\mathrm{h}\Lambda < \frac{1}{2}$. The anticipated term $M^{\mathrm{h}}_{t_n}$ is well-defined as a $\mathcal{P}_2(E)$ valued, $\mathfrak{F}_{t_n}$ measurable random variable, and $Z^{\mathrm{h}}_{t_n}$ is uniquely defined in $L^2(\mathfrak{F}_{t_n}, \mathbb{R}^q)$.*

Proof. Under Assumption 3.1, it is straightforward to prove by backward induction on n that $Y^{\mathrm{h}}_{t_n}$ is uniquely defined in $L^2(\mathfrak{F}_{t_n}, \mathbb{R})$, as a fixed point, as soon as $\mathrm{h}\Lambda < \frac{1}{2}$. Indeed, for each $0 \leq n < N$, conditioned on the knowledge of $(Y^{\mathrm{h}}_{t_m})_{m \geq n+1}, (Z^{\mathrm{h}}_{t_m})_{m \geq n}$ and X^{h}, the map

$$\chi \mapsto \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + f(t_n, X^{\mathrm{h}}_{t_n}, \chi, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h},\chi}_{t_n})\mathrm{h}_n, \text{ where}$$
$$M^{\mathrm{h},\chi}_{t_n} = M^{(\chi, Z^{\mathrm{h}}_{t_n})\mathbf{1}_{(t_n, t_{n+1}]} + \sum_{m=n+1}^{N'-1}(Y^{\mathrm{h}}_{t_m}, Z^{\mathrm{h}}_{t_m})\mathbf{1}_{(t_m, t_{m+1}]}},$$

is a contraction of $L^2(\mathfrak{F}_{t_n}, \mathbb{R})$ whenever $\Lambda \mathrm{h} < \frac{1}{2}$, as, assuming $\mathrm{h} \leq 1$,

$$\begin{aligned}
&\mathbb{E}\left[|f(t_n, X^{\mathrm{h}}_{t_n}, \chi, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h},\chi}_{t_n})\mathrm{h}_n - f(t_n, X^{\mathrm{h}}_{t_n}, \chi', Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h},\chi'}_{t_n})\mathrm{h}_n|^2\right] \\
&\leq 2\Lambda^2 \mathrm{h}_n^2 \mathbb{E}\left[|\chi - \chi'|^2 + W_2^2(M^{\mathrm{h},\chi}_{t_n}, M^{\mathrm{h},\chi'}_{t_n})\right] \\
&\leq 2\Lambda^2 \mathrm{h}^2 \mathbb{E}\left[|\chi - \chi'|^2 + \int_{t_n}^{t_{n+1}} |\chi - \chi'|^2 ds\right] \\
&\leq 4\Lambda^2 \mathrm{h}^2 \mathbb{E}\left[|\chi - \chi'|^2\right].
\end{aligned}$$

This provides the unique solvability at each backward step. Regarding $Z^{\mathrm{h}}_{t_n}$, the conditional Cauchy-Schwarz inequality yields $\mathbb{E}\left[|Z^{\mathrm{h}}_{t_n}|^2\right] \leq q\mathrm{h}_n^{-1}\mathbb{E}\left[|Y^{\mathrm{h}}_{t_{n+1}}|^2\right] < \infty$. Note also that, by immediate induction, since $(Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}) \in L^2(\mathfrak{F}_{t_n}, \mathbb{R} \times \mathbb{R}^q)$, the piecewise-constant process $\sum_{m=n}^{N'-1}(Y^{\mathrm{h}}_{t_m}, Z^{\mathrm{h}}_{t_m}, W_{t_m} - W_{t_n})\mathbf{1}_{(t_m, t_{m+1}]}$ is progressive. Hence, the anticipated term $M^{\mathrm{h}}_{t_n}$ is well-defined as an element of $\mathcal{P}_2(E)$. □

Using the Brownian martingale representation property, for each $0 \leq n < N'$, there exists a square-integrable process $\overline{Z}^{\mathrm{h}}$ such that

$$\begin{aligned} Y^{\mathrm{h}}_{t_n} &= \mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}\right] + \mathrm{h}_n f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h}}_{t_n}) \\ &= Y^{\mathrm{h}}_{t_{n+1}} + \mathrm{h}_n f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h}}_{t_n}) - \int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s dW_s. \end{aligned}$$

We then define, for each $0 \leq n < N$ and $t_n \leq t \leq t_{n+1}$,

$$Y^{\mathrm{h}}_t = Y^{\mathrm{h}}_{t_n} - f(t_n, X^{\mathrm{h}}_{t_n}, Y^{\mathrm{h}}_{t_n}, Z^{\mathrm{h}}_{t_n}, M^{\mathrm{h}}_{t_n})(t - t_n) + \int_{t_n}^{t} \overline{Z}^{\mathrm{h}}_s dW_s. \tag{34}$$

Multiplying by $(W_{t_{n+1}} - W_{t_n})$ and taking conditional expectation, we obtain by Itô isometry

$$\mathbb{E}_{t_n}\left[Y^{\mathrm{h}}_{t_{n+1}}(W_{t_{n+1}} - W_{t_n})\right] = \mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s ds\right],$$

i.e., dividing by h_n,

$$Z^{\mathrm{h}}_{t_n} = \mathrm{h}_n^{-1}\mathbb{E}_{t_n}\left[\int_{t_n}^{t_{n+1}} \overline{Z}^{\mathrm{h}}_s ds\right]. \tag{35}$$

In order to control the error between (Y, Z) and $(Y^{\mathrm{h}}, Z^{\mathrm{h}})$, we also need the following generalization of the classical discrete Gronwall's lemma.

Lemma 5.4. *Let $N \geq 1$ be an integer, $0 < h \leq 1$, $T = Nh$, $(e_n)_{n=0}^N$, $(f_n)_{n=0}^{N-1}$ and $(\varepsilon_n)_{n=0}^{N-1}$ be sequences of non-negative real numbers. Suppose there exists a constant G such that*

$$e_n + \frac{1}{2} f_n \leq (1 + Gh)\left(e_{n+1} + h^2 \sum_{k=n+1}^{N-1} e_k + h \sum_{k=n+1}^{N-1} f_k\right) + \varepsilon_n,\ 0 \leq n \leq N-1$$

Then

$$e_n + \sum_{k=0}^{N-1} f_k \leq C(G, T)\left(e_N + \sum_{k=0}^{N-1} \varepsilon_k\right),\ 0 \leq n \leq N.$$

Proof. First, define, for $0 \leq n \leq N$, $F_n = \sum_{k=n}^{N-1} f_k$ (with $F_N = 0$) and $A_n = e_n + \frac{1}{2} F_n$. Since $f_n = F_n - F_{n+1}$ and $\sum_{k=n+1}^{N-1} f_k = F_{n+1}$, we have by assumption, since $F_{n+1} =$

$2(A_{n+1} - e_{n+1}) \le 2A_{n+1}$ and $e_k \le A_k$ for all k,

$$\begin{aligned}
A_n &= e_n + \frac{1}{2} f_n + \frac{1}{2} F_{n+1} \\
&\le (1+Gh)\, e_{n+1} + \frac{1}{2}\, F_{n+1} + (1+Gh)\, h\, F_{n+1} + (1+Gh)\, h^2 \sum_{k=n+1}^{N-1} e_k + \varepsilon_n \\
&\le A_{n+1} + Ghe_{n+1} + 2(1+Gh)\, h\, (A_{n+1} - e_{n+1}) + (1+Gh)\, h^2 \sum_{k=n+1}^{N-1} A_k + \varepsilon_n \\
&\le (1+Gh)\, A_{n+1} + 2(1+Gh)\, h\, A_{n+1} + (1+Gh)\, h^2 \sum_{k=n+1}^{N-1} A_k + \varepsilon_n \\
&= (1+Gh)(1+2h) A_{n+1} + (1+Gh) h^2 \sum_{k=n+1}^{N-1} A_k + \varepsilon_n \\
&\le (1+G_1 h) A_{n+1} + G_1 h^2 \sum_{k=n+1}^{N-1} A_k + \varepsilon_n,
\end{aligned}$$

where $G_1 \ge 3G + 2$.

Defining $B_n = \max_{n \le k \le N} A_k$ (so that $B_N = A_N = e_N$), we obtain

$$\begin{aligned}
A_n &\le (1+G_1 h)\, A_{n+1} + G_1 h^2 N B_{n+1} + \varepsilon_n \\
&\le (1 + G_1 h + G_1 hT) B_{n+1} + \varepsilon_n \\
&= (1+G_2 h) B_{n+1} + \varepsilon_n,
\end{aligned}$$

where $G_2 = G_1(1+T)$. Notice that

$$\begin{aligned}
B_n = \max(A_n, B_{n+1}) &\le \max((1+G_2 h) B_{n+1} + \varepsilon_n, B_{n+1}) \\
&= (1+G_2 h) B_{n+1} + \varepsilon_n.
\end{aligned}$$

From the classical discrete Gronwall's lemma, we then obtain that

$$B_0 \le e^{G_2 T} \left(e_N + \sum_{j=0}^{N-1} \varepsilon_j \right).$$

Since $e_n \le A_n \le B_0$ and $F_0 = 2(A_0 - e_0) \le 2\, A_0 \le 2\, B_0$, we conclude

$$e_n + \sum_{k=0}^{N-1} f_k = e_n + F_0 \le 3\, B_0 \le 3\, e^{G_2 T} \left(e_N + \sum_{k=0}^{N-1} \varepsilon_k \right),$$

which gives the result for $C(G,T) = 3\, e^{G_2 T}$. □

Theorem 5.5. *Under Assumptions 3.1 and 5.1, there exists a constant $C \ge 0$ (independent of* h*) such that*

$$\sup_{t \in [0, T+\theta]} \mathbb{E}\left[|Y_t - Y_t^{\mathrm{h}}|^2 \right] + \mathbb{E}\left[\int_0^{T+\theta} |Z_t - \overline{Z}_t^{\mathrm{h}}|^2\, dt \right] \le C\, \mathrm{h}.$$

Proof. For $0 \le n < N$ and $t_n \le t < t_{n+1}$, set

$$\delta Y_t = Y_t - Y_t^{\mathrm{h}}, \qquad \delta Z_t = Z_t - \overline{Z}_t^{\mathrm{h}},$$
$$\delta f_t = f(t, X_t, Y_t, Z_t, M_t^{Y,Z}) - f(t_n, X_{t_n}^{\mathrm{h}}, Y_{t_n}^{\mathrm{h}}, Z_{t_n}^{\mathrm{h}}, M_{t_n}^{\mathrm{h}}).$$

For each $0 \le n < N$ and $t_n \le t < t_{n+1}$, applying Itô's formula to $|\delta Y_t|^2$ yields

$$\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] = \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + 2\mathbb{E}\left[\int_t^{t_{n+1}} \delta Y_s^\top \, \delta f_s \, ds\right].$$

Using Young's inequality $2ab \le \alpha a^2 + \alpha^{-1} b^2$ for a parameter $\alpha > 0$ to be chosen later,

$$\begin{aligned} &\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] \\ &\qquad \le \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \alpha \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Y_s|^2 ds\right] + \frac{1}{\alpha}\mathbb{E}\left[\int_t^{t_{n+1}} |\delta f_s|^2 ds\right]. \end{aligned} \tag{36}$$

By the Lipschitz property of f in (x, y, z, μ) and its Hölder regularity in time (see Assumptions 3.1 and 5.1(ii)), for $s \in [t_n, t_{n+1})$,

$$\begin{aligned} |\delta f_s|^2 &\le C\Big(|s - t_n| + |X_s - X_{t_n}^{\mathrm{h}}|^2 + |Y_s - Y_{t_n}^{\mathrm{h}}|^2 + |Z_s - Z_{t_n}^{\mathrm{h}}|^2 + W_2^2(M_s^{Y,Z}, M_{t_n}^{\mathrm{h}})\Big) \\ &\le C\Big(\mathrm{h} + |X_s - X_{t_n}^{\mathrm{h}}|^2 + |Y_s - Y_{t_n}|^2 + |\delta Y_{t_n}|^2 \\ &\qquad + |Z_s - \widetilde{Z}_{t_n}^{\mathrm{h}}|^2 + |\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}}|^2 \\ &\qquad + W_2^2(M_s^{Y,Z}, M_{t_n}^{Y,Z}) + W_2^2(M_{t_n}^{Y,Z}, M_{t_n}^{\mathrm{h}})\Big), \end{aligned}$$

where $\widetilde{Z}_{t_n}^{\mathrm{h}}$ is defined in (30).

For any $t_n \le s \le t_{n+1}$, we have, from (31) and from Theorem 3.6, for some constant $C \ge 0$,

$$\begin{aligned} \mathbb{E}\left[|X_s - X_{t_n}^{\mathrm{h}}|^2\right] &\le 2\mathbb{E}\left[|X_s - X_{t_n}|^2\right] + 2\mathbb{E}\left[|X_{t_n} - X_{t_n}^{\mathrm{h}}|^2\right] \le C\mathrm{h}, \\ \mathbb{E}\left[|Y_s - Y_{t_n}|^2\right] &= \mathbb{E}\left[|u(s, X_s) - u(t_n, X_{t_n})|^2\right] \le C\mathrm{h}. \end{aligned}$$

In addition, from (35) and the definition of $\widetilde{Z}_{t_n}^{\mathrm{h}}$, by Jensen's inequality, we get

$$\mathbb{E}\left[|\widetilde{Z}_{t_n}^{\mathrm{h}} - Z_{t_n}^{\mathrm{h}}|^2\right] \le \mathbb{E}\left[\left|\mathbb{E}_{t_n}\left[\mathrm{h}_n^{-1}\int_{t_n}^{t_{n+1}} \delta Z_s ds\right]\right|^2\right] \le \mathrm{h}_n^{-1}\mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right]. \tag{37}$$

Last, the anticipated terms are controlled by Assumption 5.1(iii) and (32) as follows. For the first bound, by Assumption 5.1(iii),

$$\mathbb{E}\left[W_2^2(M_s^{Y,Z}, M_{t_n}^{Y,Z})\right] \le \Lambda \mathrm{h}.$$

For the second bound, we use (32) to obtain

$$
\begin{aligned}
&\mathbb{E}\left[W_2^2(M_{t_n}^{Y,Z}, M_{t_n}^{\mathrm{h}})\right] \\
&\leq \sum_{k=n}^{N'-1} \mathbb{E}\left[\int_{t_k}^{t_{k+1}} (|Y_r - Y_{t_k}^{\mathrm{h}}|^2 + |Z_r - Z_{t_k}^{\mathrm{h}}|^2 + |W_r - W_{t_k}|^2) dr\right] \\
&\leq 2\sum_{k=n}^{N'-1} \int_{t_k}^{t_{k+1}} (\mathbb{E}\left[|Y_r - Y_{t_k}|^2\right] + \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathbb{E}\left[|Z_r - \widetilde{Z}_{t_k}^{\mathrm{h}}|^2\right] + \mathbb{E}\left[|\widetilde{Z}_{t_k}^{\mathrm{h}} - Z_{t_k}^{\mathrm{h}}|^2\right] + \mathrm{h}) dr \\
&\leq C\sum_{k=n}^{N'-1} \left(\mathrm{h}^2 + \mathrm{h}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|Z_r - \widetilde{Z}_{t_k}^{\mathrm{h}}|^2\right] dr + \mathrm{h}\mathbb{E}\left[|\widetilde{Z}_{t_k}^{\mathrm{h}} - Z_{t_k}^{\mathrm{h}}|^2\right]\right) \\
&\leq C\mathrm{h} + \sum_{k=n}^{N'-1} \left(\mathrm{h}\mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\right) \\
&\leq C\mathrm{h} + \mathrm{h}\sum_{k=n}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds,
\end{aligned}
$$

where we used (37), Lemma 5.1 above, $\sum_{k=N}^{N'-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] \leq C$ and $\int_T^{T+\theta} \mathbb{E}\left[|\delta Z_s|^2\right] ds \leq C\mathrm{h}$, by Lemma 5.2.

All in all, we have shown that, for any $t_n \leq s \leq t_{n+1}$,

$$
\begin{aligned}
\mathbb{E}\left[|\delta f_s|^2\right] \leq & C\Big(\mathrm{h} + \mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \mathbb{E}\left[|Z_s - \widetilde{Z}_{t_n}^{\mathrm{h}}|^2\right] + \mathrm{h}_n^{-1}\mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right] \\
& + \mathrm{h}\sum_{k=n}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\Big),
\end{aligned}
$$

hence, for any $t_n \leq t < t_{n+1}$,

$$
\begin{aligned}
\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] \leq & \mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \alpha\mathbb{E}\left[\int_t^{t_{n+1}} |\delta Y_s|^2 ds\right] \\
& + \frac{C}{\alpha}\Big(\mathrm{h}^2 + \mathrm{h}\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}_{t_n}^{\mathrm{h}}|^2\right] ds \\
& + \mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right] + \mathrm{h}^2\sum_{k=n}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] \\
& + \mathrm{h}\sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\Big).
\end{aligned}
$$

Using Gronwall's lemma on the interval $[t_n, t_{n+1}]$, we obtain, for each $t_n \leq t \leq t_{n+1}$,

$$
\begin{aligned}
\mathbb{E}\left[|\delta Y_t|^2\right] \leq e^{\alpha\mathrm{h}}\Big(&\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \frac{C}{\alpha}\Big(\mathrm{h}^2 + \mathrm{h}\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}_{t_n}^{\mathrm{h}}|^2\right] ds \quad (38) \\
& + \mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right] + \mathrm{h}^2\sum_{k=n}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathrm{h}\sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\Big)\Big),
\end{aligned}
$$

and substituting this estimate back into the previous inequality to get

$$\begin{aligned}
&\mathbb{E}\left[|\delta Y_t|^2\right] + \mathbb{E}\left[\int_t^{t_{n+1}} |\delta Z_s|^2 ds\right] \\
\leq &(1+\alpha \mathrm{h} e^{\alpha \mathrm{h}})\Big(\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \frac{C}{\alpha}\Big(\mathrm{h}^2 + \mathrm{h}\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] ds \\
&+ \mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right] + \mathrm{h}^2 \sum_{k=n}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathrm{h} \sum_{k=n}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\Big)\Big). \quad (39)
\end{aligned}$$

For h small enough and α large enough, we obtain, for $t = t_n$,

$$\begin{aligned}
\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \frac{1}{2}\mathbb{E}\left[\int_{t_n}^{t_{n+1}} |\delta Z_s|^2 ds\right] \leq &(1+C\mathrm{h})\Big(\mathbb{E}\left[|\delta Y_{t_{n+1}}|^2\right] + \mathrm{h}^2 + \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] ds \\
&+ \mathrm{h}^2 \sum_{k=n+1}^{N-1} \mathbb{E}\left[|\delta Y_{t_k}|^2\right] + \mathrm{h} \sum_{k=n+1}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds\Big).
\end{aligned}$$

Using the extended Gronwall lemma 5.4 with $h = \mathrm{h}$, $e_n = \mathbb{E}\left[|\delta Y_{t_n}|^2\right]$, $f_n = \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|\delta Z_s|^2\right] ds$ and $\varepsilon_n = (1+C\mathrm{h})(\mathrm{h}^2 + \int_{t_n}^{t_{n+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_n}|^2\right] ds)$ then yields

$$\begin{aligned}
\mathbb{E}\left[|\delta Y_{t_n}|^2\right] + \int_0^T \mathbb{E}\left[|\delta Z_s|^2\right] ds &\leq C\Big(\mathbb{E}\left[|\delta Y_T|^2\right] + (1+C\mathrm{h}) \sum_{k=0}^{N-1} (\mathrm{h}^2 + \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_k}|^2\right] ds)\Big) \\
&\leq C\Big(C\mathrm{h} + C(1+C\mathrm{h})\mathrm{h}\Big) \leq C\mathrm{h}, \quad (40)
\end{aligned}$$

where we used Lemma 5.2 to bound $\mathbb{E}\left[|\delta Y_T|^2\right]$ by $C\mathrm{h}$, and Lemma 5.1 to bound $\sum_{k=0}^{N-1} \int_{t_k}^{t_{k+1}} \mathbb{E}\left[|Z_s - \widetilde{Z}^{\mathrm{h}}_{t_k}|^2\right] ds$ by $C\mathrm{h}$.

To conclude, coming back to (38) and using the above estimate, we obtain, for each $t \in [0,T]$,

$$\mathbb{E}\left[|\delta Y_t|^2\right] \leq C\mathrm{h}.$$

Since we also showed on our way that $\int_0^T \mathbb{E}\left[|\delta Z_s|^2\right] ds \leq C\mathrm{h}$ and $\int_T^{T+\theta} \mathbb{E}\left[|\delta Z_s|^2\right] ds \leq C\mathrm{h}$, we are done. □

6 Numerical results

6.1 Local Hermite regression scheme

Let

$$\mathrm{h} = \{0 = t_0 < \cdots < t_N = T < t_{N+1} < \cdots < t_{N'} = T + \delta\}$$

be a uniform grid with time step $h = t_{n+1} - t_n$, chosen so that δ/h is an integer. We simulate M independent Brownian paths $(W_{t_n})_{0 \leq n \leq N'}$. On the terminal interval $[T, T+\delta]$, the prescribed processes are known and we set

$$Y^{\mathrm{h}}_{t_n} = \Phi(t_n, W_{t_n}), \qquad Z^{\mathrm{h}}_{t_n} = \Psi(t_n, W_{t_n}), \qquad N \leq n \leq N'.$$

These terminal data are the only oracle values used by the backward algorithm. All four quantities of the numerical scheme introduced in Section 5 are approximated with the same degree-three Hermite model

$$H_0(x) = 1, \qquad H_1(x) = x, \qquad H_2(x) = \frac{x^2 - 1}{\sqrt{2}}, \qquad H_3(x) = \frac{x^3 - 3x}{\sqrt{6}}.$$

For $\beta = (\beta_0, \dots, \beta_3) \in \mathbb{R}^4$, write

$$P_\beta(x) = \sum_{k=0}^{3} \beta_k H_k(x).$$

For $n = N-1, \dots, 0$, all regressors at dates later than t_n are frozen and four regressions are performed. Let M_n denote the anticipated term constructed from the regressors already available at that stage of the backward recursion. The four parameter vectors are determined sequentially. First,

$$\widehat{\beta}_n^{\widetilde{Y}} = \arg\min_{\beta \in \mathbb{R}^4} \mathbb{E}\left[\left| P_\beta(X_n) - Y^{\mathrm{h}}_{t_{n+1}} \right|^2 \right], \qquad \widehat{\widetilde{Y}}_n = P_{\widehat{\beta}_n^{\widetilde{Y}}}(X_n).$$

$$\widehat{\beta}_n^{Z} = \arg\min_{\beta \in \mathbb{R}^4} \mathbb{E}\left[\left| P_\beta(X_n) - \frac{\big(Y^{\mathrm{h}}_{t_{n+1}} - \widehat{\widetilde{Y}}_n\big)\mathrm{h}W_n}{h} \right|^2 \right], \qquad \widehat{Z}_n = P_{\widehat{\beta}_n^{Z}}(X_n),$$

$$\widehat{\beta}_n^{A} = \arg\min_{\beta \in \mathbb{R}^4} \mathbb{E}\Big[|P_\beta(X_n) - M_n|^2 \Big], \qquad \widehat{M}_n = P_{\widehat{\beta}_n^{M}}(X_n),$$

$$\widehat{\beta}_n^{Y} = \arg\min_{\beta \in \mathbb{R}^4} \mathbb{E}\left[\left| P_\beta(X_n) - \left(\widehat{\widetilde{Y}}_n + hf\Big(t_n, P_{\widehat{\beta}_n^{M}}(X_n)\Big) \right) \right|^2 \right], \qquad \widehat{Y}_n = P_{\widehat{\beta}_n^{Y}}(X_n).$$

All four problems are instances of the same ridge regression. For a generic sample target $(R_n^{(m)})_{1 \le m \le M}$, we compute

$$\widehat{\beta}_n[R] = \arg\min_{\beta \in \mathbb{R}^4} \left\{ \sum_{m=1}^{M} \left| P_\beta(X_n^{(m)}) - R_n^{(m)} \right|^2 + \lambda |\beta|^2 \right\}, \qquad \lambda = 10^{-10}.$$

The normal equations are solved in double precision. The small ridge parameter is used only to stabilize the linear system.

6.2 Benchmark examples

We test the regression scheme on three explicit one-dimensional examples adapted from Peng and Yang (2009). In each case, the closed-form solution is used only for out-of-sample evaluation.

Example 4.3 of Peng and Yang (2009). For $0 \le t \le T$,

$$Y_t = TW_T - \int_t^T \frac{1}{s+\delta}\, \mathbb{E}_s[Y_{s+\delta}]\, ds - \int_t^T Z_s\, dW_s,$$

with $Y_t = tW_t$ on $[T, T+\delta]$. The exact solution is

$$Y_t = tW_t, \qquad Z_t = t, \qquad 0 \le t \le T + \delta.$$

Example 5.3 (I) of Peng and Yang (2009). For $0 \le t \le T$,

$$Y_t = W_T^2 - T - \int_t^T \sqrt{\frac{\mathrm{h}}{2\delta}}\, \mathbb{E}_s[|Z_{s+\delta} - Z_s|]\, ds - \int_t^T Z_s\, dW_s,$$

with

$$Y_t = W_t^2 - T - (T-t), \qquad Z_t = 2W_t, \qquad t \in [T, T+\delta].$$

The exact solution, on the full interval, is

$$Y_t = W_t^2 - T - (T-t), \qquad Z_t = 2W_t.$$

Example 5.3 (II) of Peng and Yang (2009). For $0 \le t \le T$,

$$Y_t = 4W_T^2 - \int_t^T \sqrt{\frac{\mathrm{h}}{2\delta}}\, \mathbb{E}_s[|Z_{s+\delta} - Z_s|]\, ds - \int_t^T Z_s\, dW_s,$$

with

$$Y_t = 4W_t^2 - 4(T-t), \qquad Z_t = 8W_t, \qquad t \in [T, T+\delta].$$

The exact solution is

$$Y_t = 4W_t^2 - 4(T-t), \qquad Z_t = 8W_t.$$

Figures 1, 3, and 5 compare the 5%, 50%, and 95% quantiles of the learned processes with their exact counterparts. Solid lines denote exact quantiles and hollow circles denote Hermite regression estimates. The markers lie on the corresponding lines in all three examples. In particular, the method recovers the deterministic process $Z_t = t$ in Example 4.3 even though the regressions are functions of the Brownian state.

The quantile figures use a representative model trained with $N = 20$ and 10^6 Brownian trajectories; the displayed quantiles are computed from an independent sample of 50 000 paths. All reported training runs use the same degree-three basis and the same ridge penalty.

To assess the estimate of Theorem 5.5, we also consider

$$N \in \{5, 10, 20, 40, 80, 160\},$$

or equivalently,

$$h \in \{0.2, 0.1, 0.05, 0.025, 0.0125, 0.00625\}.$$

The discrete processes are reconstructed as left-constant processes on eight substeps per grid interval. Figures 2, 4, and 6 report separately

$$\max_{0 \le t \le T} \mathbb{E}|Y_t - \bar{Y}_t^h|^2 \qquad \text{and} \qquad \mathbb{E}\int_0^T |Z_t - \bar{Z}_t^h|^2\, dt.$$

Each point is averaged over three independent training seeds and evaluated on 100 000 independent paths. For Examples 5.3 (I) and 5.3 (II), both errors have empirical log–log slopes close to one. In Example 4.3, the Y error has slope close to one, whereas the Z error decreases faster, with a slope close to two over most of the retained range. These results are consistent with the $O(h)$ squared-error estimate of Theorem 5.5; the faster decay observed for Z in Example 4.3 is specific to this benchmark.

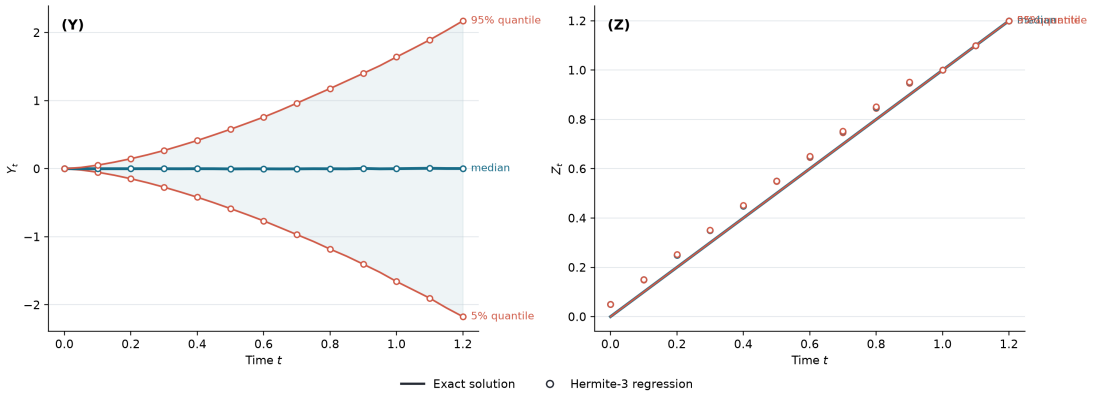


Figure 1: Example 4.3: exact quantiles (solid lines) and Hermite-3 regression quantiles (hollow circles) for Y and Z.

7 Conclusion

In this paper, we have introduced a new class of anticipated backward stochastic differential equations (ABSDEs) in a Brownian filtration, and we have proved existence and uniqueness of their solution under classical Lipschitz assumptions, in a non-Markovian framework. A comparison theorem is proved in the case where the anticipated term does not depend on the Z component of the solution. In a Markovian framework, we obtain the representation of the solution (Y, Z) as a deterministic function of the time and of the Markovian state. We have introduced an approximation scheme for Markovian ABSDEs and proved its convergence, and we proved that its solution can be represented in terms of a viscosity solution to a nonlocal (in time and space) PDE.

The study of the PDE associated to Markovian ABSDEs, in particular the important question of uniqueness of a viscosity solution and the regularity of the solution under suitable assumptions on the coefficients, is left for further research. The uniqueness issue is quite delicate already in the classical semilinear case, relying on Ishii's lemma (see e.g. Crandall et al. (1992)). In the present setting, the PDE (27) is nonlocal in time: the equation at (t, x) involves the values of the solution on the entire future domain $[t, T + \theta] \times \mathbb{R}^d$, creating an obstruction to the classical doubling-of-variables technique used to prove comparison principles for viscosity solutions. A possible strategy to overcome this difficulty would be to work backward in time on intervals of size at most δ, assuming the anticipated term at any time t introduces a dependence only upon the future interval $[t, t+\theta]$, assuming also that the anticipated term induces only a small perturbation from a classical non-anticipated driver. This analysis is beyond the scope of the present paper and will be addressed in future work.

In addition, the representation of the ABSDE in terms of the viscosity solution to the nonlocal PDE relies on the comparison theorem, hence is restricted to the case where the anticipated term does not depend on Z. Proving a comparison theorem with the anticipated term depending somehow on the Z process remains an open problem. Note however that the dependence on Z cannot be arbitrary: in the special case of Example 2.4, if Assumption 2.2 (resp. 2.3) is not satisfied, then the ABSDE may fail

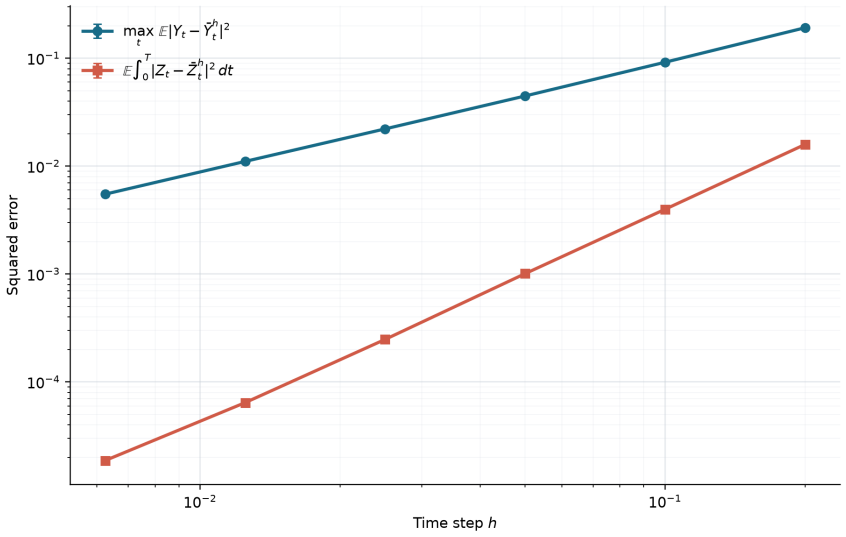


Figure 2: Example 4.3: the two time-reconstruction errors in Theorem 5.5.

to satisfy a comparison principle, as shown by Peng and Yang (2009, Example 5.3 (resp. Example 5.2)).

Beyond the PDE issue already mentioned, this work opens several BSDE perspectives. First, it would be interesting to relax the Lipschitz assumptions on the coefficients, and in particular to consider quadratic growth in Z. Second, it would be interesting to extend our results to the case of anticipated ABSDEs also driven by a jump process. Last, it would be interesting to apply our results to other problems in economics and finance (beyond the XVA setup of Abbas-Turki et al. (2025)), such as optimal control problems with delay.

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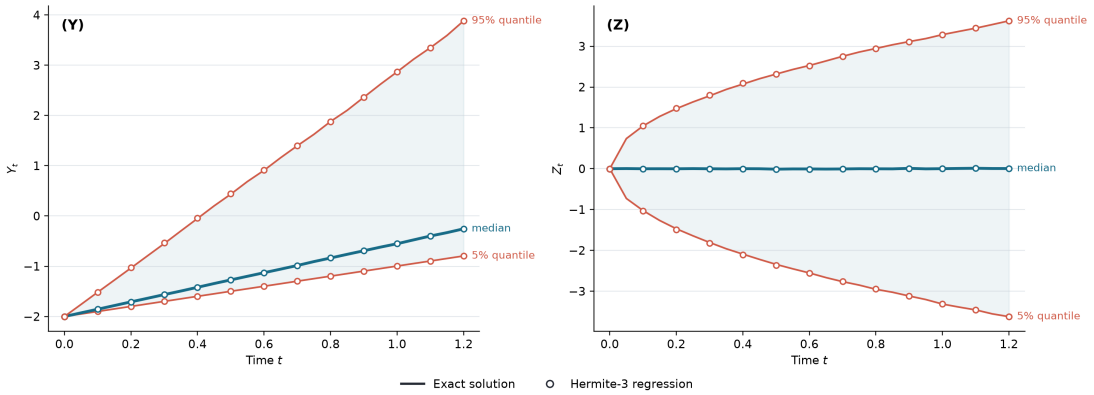


Figure 3: Example 5.3 (I): exact quantiles (solid lines) and Hermite-3 regression quantiles (hollow circles) for Y and Z.

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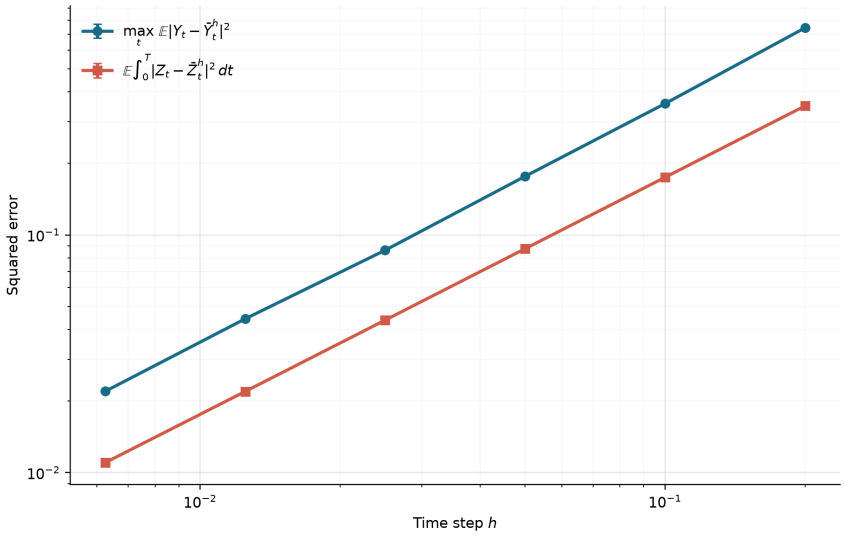


Figure 4: Example 5.3 (I): the two time-reconstruction errors in Theorem 5.5.

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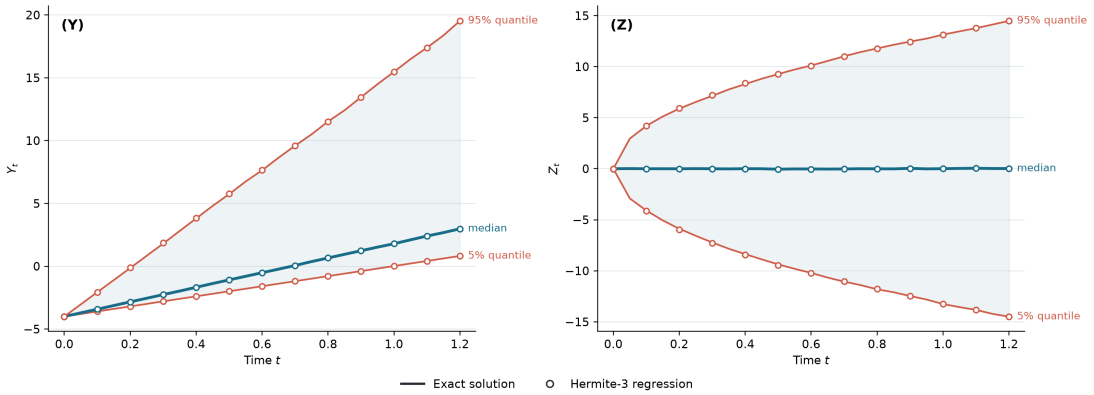


Figure 5: Example 5.3 (II): exact quantiles (solid lines) and Hermite-3 regression quantiles (hollow circles) for Y and Z.

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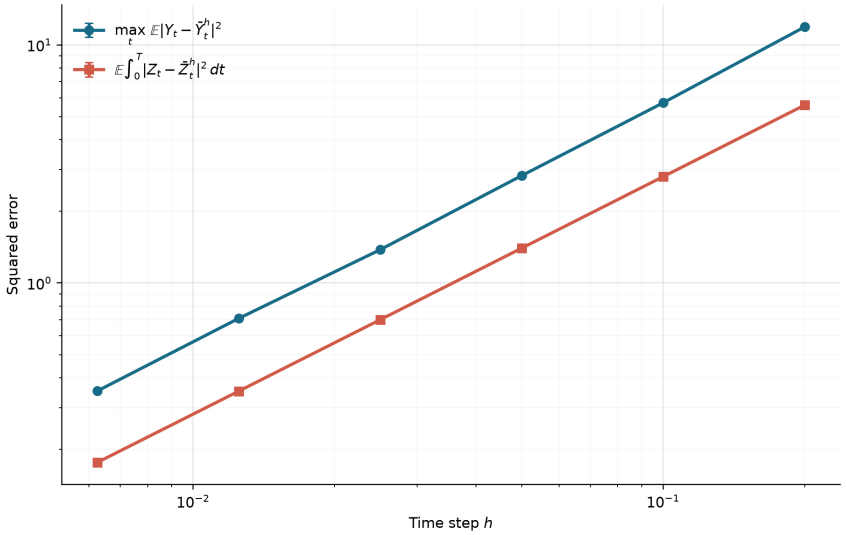


Figure 6: Example 5.3 (II): the two time-reconstruction errors in Theorem 5.5.